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(54) **POLYPEPTIDES HAVING XYLANASE
ACTIVITY AND POLYNUCLEOTIDES
ENCODING SAME**

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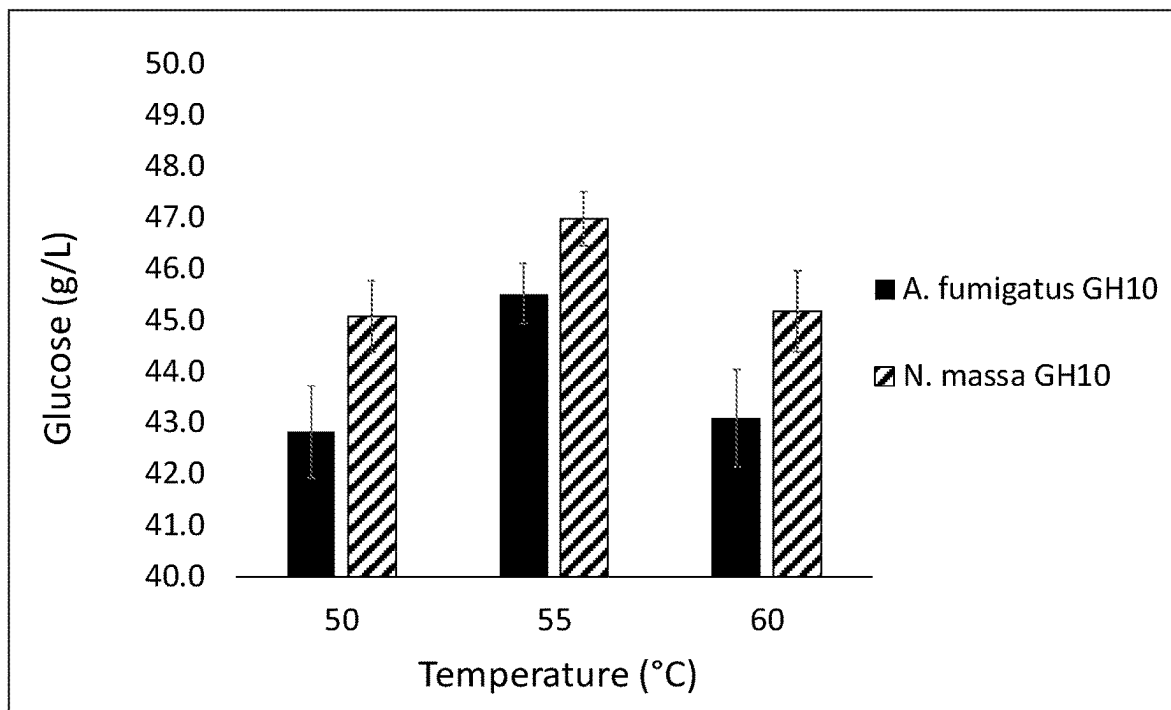
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(57) **ABSTRACT**

The present invention relates to isolated polypeptides having
xylanase activity, catalytic domains, carbohydrate binding
modules and polynucleotides encoding the polypeptides,
catalytic domains or carbohydrate binding modules. The
invention also relates to nucleic acid constructs, vectors, and
host cells comprising the polynucleotides as well as methods
of producing and using the polypeptides, catalytic domains
or carbohydrate binding modules.

16 Claims, 1 Drawing Sheet

Specification includes a Sequence Listing.



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POLYPEPTIDES HAVING XYLANASE ACTIVITY AND POLYNUCLEOTIDES ENCODING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a 35 U.S.C. 371 national application of international application no. PCT/US2019/065401 filed Dec. 10, 2019, which claims priority or the benefit under 35 U.S.C. 119 of U.S. provisional application No. 62/778,615 filed Dec. 12, 2018, the contents of which are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING

This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to polypeptides having xylanase activity, catalytic domains, and carbohydrate binding modules, and polynucleotides encoding the polypeptides, catalytic domains, and carbohydrate binding modules. The invention also relates to nucleic acid constructs, vectors, and host cells comprising the polynucleotides as well as methods of producing and using the polypeptides, catalytic domains, and carbohydrate binding modules.

Description of the Related Art

Lignocellulose, the world's largest renewable biomass resource, is composed mainly of lignin, cellulose, and hemicellulose. Cellulose is a polymer of glucose linked by beta-1,4-bonds known as beta-linked glucans. Hemicellulose is composed of xylans, which are polysaccharides formed from 1,4-beta-glycoside-linked D-xylopyranoses.

Many microorganisms produce enzymes that hydrolyze the beta-linked glucans and xylans. Endoglucanases digest the cellulose polymer at random locations, opening it to attack by cellobiohydrolases. Cellobiohydrolases sequentially release molecules of cellobiose from the ends of the cellulose polymer. Cellobiose is a water-soluble beta-1,4-linked dimer of glucose. Beta-glucosidases hydrolyze cellobiose to glucose. Xylanases (e.g., endo-1,4-beta-xylanase, EC 3.2.1.8) hydrolyze internal beta-1,4-xylosidic linkages in xylan to produce smaller molecular weight xylose and xylo-oligomers. Beta-xylosidases catalyze the exo-hydrolysis of short beta (1-4)-xylooligosaccharides to remove successive D-xylose residues from non-reducing termini.

The conversion of lignocellulosic feedstocks into ethanol has the advantages of the availability of large amounts of feedstock, the desirability of avoiding burning or land filling the materials, and the cleanliness of the ethanol fuel. Wood, agricultural residues, herbaceous crops, and municipal solid wastes have been considered as feedstocks for ethanol production. These materials primarily consist of cellulose, hemicellulose, and lignin. Once the cellulose is converted to glucose and the hemicellulose to xylose, the glucose and xylose are fermented by yeast into ethanol.

There is a need in the art for new enzymes that can deconstruct cellulosic or hemicellulosic material more efficiently and provide cost-effective enzyme solutions for saccharification of cellulosic material.

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The present invention provides polypeptides having xylanase activity and polynucleotides thereof and processes of producing and using the polypeptides having xylanase activity.

SUMMARY OF THE INVENTION

The present invention relates to isolated polypeptides having xylanase activity selected from the group consisting of:

- (a) a polypeptide having at least 99% sequence identity to the mature polypeptide of SEQ ID NO: 2;
- (b) a polypeptide encoded by a polynucleotide having at least 99% sequence identity to the mature polypeptide coding sequence of SEQ ID NO: 1;
- (c) a variant of the mature polypeptide of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and
- (d) a fragment of the polypeptide of (a), (b), or (c) that has xylanase activity.

The present invention also relates to isolated polypeptides comprising a catalytic domain selected from the group consisting of:

- (a) a catalytic domain having at least 99% sequence identity to amino acids 20 to 341 of SEQ ID NO: 2;
- (b) a catalytic domain encoded by a polynucleotide having at least 99% sequence identity to nucleotides 108 to 1241 of SEQ ID NO: 1;
- (c) a variant of amino acids 20 to 341 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and
- (d) a fragment of the catalytic domain of (a), (b), or (c) that has xylanase activity.

The present invention also relates to isolated polypeptides comprising a carbohydrate binding module selected from the group consisting of:

- (a) a carbohydrate binding module having at least 99% sequence identity to amino acids 363 to 399 of SEQ ID NO: 2;
- (b) a carbohydrate binding module encoded by a polynucleotide having at least 99% sequence identity to nucleotides 1305 to 1415 of SEQ ID NO: 1;
- (c) a variant of amino acids 363 to 399 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and
- (d) a fragment of the carbohydrate binding module of (a), (b), or (c) that has binding activity.

The present invention also relates to isolated polynucleotides encoding the polypeptides of the present invention; nucleic acid constructs, recombinant expression vectors, and recombinant host cells comprising the polynucleotides; and methods of producing the polypeptides.

The present invention also relates to processes for degrading a cellulosic or hemicellulosic material, comprising: treating the cellulosic or hemicellulosic material with an enzyme composition comprising such a polypeptide having xylanase activity.

The present invention also relates to processes for producing a fermentation product, comprising:

- (a) saccharifying a cellulosic or hemicellulosic material with an enzyme composition comprising such a polypeptide having xylanase activity;
- (b) fermenting the saccharified cellulosic or hemicellulosic material with one or more fermenting microorganisms to produce the fermentation product; and
- (c) recovering the fermentation product from the fermentation.

The present invention also relates to processes of fermenting a cellulosic or hemicellulosic material, comprising: fermenting the cellulosic or hemicellulosic material with one or more fermenting microorganisms, wherein the cellulosic or hemicellulosic material is saccharified with an enzyme composition comprising such a polypeptide having xylanase activity.

The present invention also relates to an isolated polynucleotide encoding a signal peptide comprising or consisting of amino acids 1 to 19 of SEQ ID NO: 2, which is operably linked to a gene encoding a protein, wherein the gene is foreign to the polynucleotide encoding the signal peptide; nucleic acid constructs, expression vectors, and recombinant host cells comprising such a polynucleotide; and methods of producing a protein.

BRIEF DESCRIPTION OF THE FIGURE

FIG. 1 shows the effect of adding *Neosartorya massa* GH10 xylanase to a cellulolytic enzyme composition without GH10 compared to an *Aspergillus fumigatus* GH10 xylanase at 50° C., 55° C., and 60° C.

DEFINITIONS

Reference to “about” a value or parameter herein includes aspects that are directed to that value or parameter per se. For example, description referring to “about X” includes the aspect “X”.

As used herein and in the appended claims, the singular forms “a,” “or,” and “the” include plural referents unless the context clearly dictates otherwise. It is understood that the aspects of the invention described herein include “consisting” and/or “consisting essentially of” aspects.

Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.

Acetylxyylan esterase: The term “acetylxyylan esterase” means a carboxylesterase (EC 3.1.1.72) that catalyzes the hydrolysis of acetyl groups from polymeric xylan, acetylated xylose, acetylated glucose, alpha-naphthyl acetate, and p-nitrophenyl acetate. Acetylxyylan esterase activity can be determined using 0.5 mM p-nitrophenylacetate as substrate in 50 mM sodium acetate pH 5.0 containing 0.01% TWEEN™ 20 (polyoxyethylene sorbitan monolaurate). One unit of acetylxyylan esterase is defined as the amount of enzyme capable of releasing 1 μmole of p-nitrophenolate anion per minute at pH 5, 25° C.

Allelic variant: The term “allelic variant” means any of two or more alternative forms of a gene occupying the same chromosomal locus. Allelic variation arises naturally through mutation, and may result in polymorphism within populations. Gene mutations can be silent (no change in the encoded polypeptide) or may encode polypeptides having altered amino acid sequences. An allelic variant of a polypeptide is a polypeptide encoded by an allelic variant of a gene.

Alpha-L-arabinofuranosidase: The term “alpha-L-arabinofuranosidase” means an alpha-L-arabinofuranoside arabinofuranohydrolase (EC 3.2.1.55) that catalyzes the hydrolysis of terminal non-reducing alpha-L-arabinofuranoside residues in alpha-L-arabinosides. The enzyme acts on alpha-L-arabinofuranosides, alpha-L-arabinans containing (1,3)- and/or (1,5)-linkages, arabinoxylans, and arabinogalactans. Alpha-L-arabinofuranosidase is also known as arabinosidase, alpha-arabinosidase, alpha-L-arabinosidase, alpha-ara-

binofuranosidase, polysaccharide alpha-L-arabinofuranosidase, alpha-L-arabinofuranoside hydrolase, L-arabinosidase, or alpha-L-arabinanase. Alpha-L-arabinofuranosidase activity can be determined using 5 mg of medium viscosity wheat arabinoxylan (Megazyme International Ireland, Ltd.) per ml of 100 mM sodium acetate pH 5 in a total volume of 200 μl for 30 minutes at 40° C. followed by arabinose analysis by AMINEX® HPX-87H column chromatography (Bio-Rad Laboratories, Inc.).

Alpha-glucuronidase: The term “alpha-glucuronidase” means an alpha-D-glucosiduronate glucuronohydrolase (EC 3.2.1.139) that catalyzes the hydrolysis of an alpha-D-glucuronoside to D-glucuronate and an alcohol. Alpha-glucuronidase activity can be determined according to de Vries, 1998, J. Bacteriol. 180: 243-249. One unit of alpha-glucuronidase equals the amount of enzyme capable of releasing 1 μmole of glucuronic or 4-O-methylglucuronic acid per minute at pH 5, 40° C.

Auxiliary Activity 9 polypeptide: The term “Auxiliary Activity 9 polypeptide” or “AA9 polypeptide” means a polypeptide classified as a lytic polysaccharide monooxygenase (Quinlan et al., 2011, *Proc. Natl. Acad. Sci. USA* 108: 15079-15084; Phillips et al., 2011, *ACS Chem. Biol.* 6: 1399-1406; Li et al., 2012, *Structure* 20: 1051-1061). AA9 polypeptides were formerly classified into the glycoside hydrolase Family 61 (GH61) according to Henrissat, 1991, *Biochem. J.* 280: 309-316, and Henrissat and Bairoch, 1996, *Biochem. J.* 316: 695-696.

AA9 polypeptides enhance the hydrolysis of a cellulosic material by an enzyme having cellulolytic activity. Cellulolytic enhancing activity can be determined by measuring the increase in reducing sugars or the increase of the total of cellobiose and glucose from the hydrolysis of a cellulosic material by cellulolytic enzyme under the following conditions: 1-50 mg of total protein/g of cellulose in pretreated corn stover (PCS), wherein total protein is comprised of 50-99.5% w/w cellulolytic enzyme protein and 0.5-50% w/w protein of an AA9 polypeptide for 1-7 days at a suitable temperature, such as 40° C.-80° C., e.g., 40° C., 45° C., 50° C., 55° C., 60° C., 65° C., 70° C., 75° C., or 80° C. and a suitable pH, such as 4-9, e.g., 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, or 9.0, compared to a control hydrolysis with equal total protein loading without cellulolytic enhancing activity (1-50 mg of cellulolytic protein/g of cellulose in PCS).

AA9 polypeptide enhancing activity can be determined using a mixture of CELLUCLAST™ 1.5 L (Novozymes A/S, Bagsvrd, Denmark) and beta-glucosidase as the source of the cellulolytic activity, wherein the beta-glucosidase is present at a weight of at least 2-5% protein of the cellulase protein loading. In one aspect, the beta-glucosidase is an *Aspergillus oryzae* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* according to WO 02/095014). In another aspect, the beta-glucosidase is an *Aspergillus fumigatus* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* as described in WO 02/095014).

AA9 polypeptide enhancing activity can also be determined by incubating an AA9 polypeptide with 0.5% phosphoric acid swollen cellulose (PASC), 100 mM sodium acetate pH 5, 1 mM MnSO₄, 0.1% gallic acid, 0.025 mg/ml of *Aspergillus fumigatus* beta-glucosidase, and 0.01% TRITON® X-100 (4-(1,1,3,3-tetramethylbutyl)phenyl-polyethylene glycol) for 24-96 hours at 40° C. followed by determination of the glucose released from the PASC. AA9

polypeptide enhancing activity can also be determined according to WO 2013/028928 for high temperature compositions.

AA9 polypeptides enhance the hydrolysis of a cellulosic material catalyzed by enzyme having cellulolytic activity by reducing the amount of cellulolytic enzyme required to reach the same degree of hydrolysis preferably at least 1.01-fold, e.g., at least 1.05-fold, at least 1.10-fold, at least 1.25-fold, at least 1.5-fold, at least 2-fold, at least 3-fold, at least 4-fold, at least 5-fold, at least 10-fold, or at least 20-fold.

The AA9 polypeptide can be used in the presence of a soluble activating divalent metal cation according to WO 2008/151043 or WO 2012/122518, e.g., manganese or copper.

The AA9 polypeptide can also be used in the presence of a dioxy compound, a bicyclic compound, a heterocyclic compound, a nitrogen-containing compound, a quinone compound, a sulfur-containing compound, or a liquor obtained from a pretreated cellulosic or hemicellulosic material such as pretreated corn stover (WO 2012/021394, WO 2012/021395, WO 2012/021396, WO 2012/021399, WO 2012/021400, WO 2012/021401, WO 2012/021408, and WO 2012/021410).

Beta-glucosidase: The term "beta-glucosidase" means a beta-D-glucoside glucohydrolase (E.C. 3.2.1.21) that catalyzes the hydrolysis of terminal non-reducing beta-D-glucose residues with the release of beta-D-glucose. Beta-glucosidase activity can be determined using p-nitrophenyl-beta-D-glucopyranoside as substrate according to the procedure of Venturi et al., 2002, *J. Basic Microbiol.* 42: 55-66. One unit of beta-glucosidase is defined as 1.0 μ mole of p-nitrophenolate anion produced per minute at 25° C., pH 4.8 from 1 mM p-nitrophenyl-beta-D-glucopyranoside as substrate in 50 mM sodium citrate containing 0.01% TWEEN® 20.

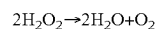
Beta-xylosidase: The term "beta-xylosidase" means a beta-D-xyloside xylohydrolase (E.C. 3.2.1.37) that catalyzes the exo-hydrolysis of short beta (1 $\rightarrow$ 4)-xylooligosaccharides to remove successive D-xylose residues from non-reducing termini. Beta-xylosidase activity can be determined using 1 mM p-nitrophenyl-beta-D-xyloside as substrate in 100 mM sodium citrate containing 0.01% TWEEN® 20 at pH 5, 40° C. One unit of beta-xylosidase is defined as 1.0 μ mole of p-nitrophenolate anion produced per minute at 40° C., pH 5 from 1 mM p-nitrophenyl-beta-D-xyloside in 100 mM sodium citrate containing 0.01% TWEEN® 20.

cDNA: The term "cDNA" means a DNA molecule that can be prepared by reverse transcription from a mature, spliced, mRNA molecule obtained from a eukaryotic or prokaryotic cell. cDNA lacks intron sequences that may be present in the corresponding genomic DNA. The initial, primary RNA transcript is a precursor to mRNA that is processed through a series of steps, including splicing, before appearing as mature spliced mRNA.

Carbohydrate binding module: The term "carbohydrate binding module" means a domain within a carbohydrate-active enzyme that provides carbohydrate-binding activity (Boraston et al., 2004, *Biochem. J.* 383: 769-781). A majority of known carbohydrate binding modules (CBMs) are contiguous amino acid sequences with a discrete fold. The carbohydrate binding module (CBM) is typically found either at the N-terminal or at the C-terminal extremity of an enzyme. Some CBMs are known to have specificity for cellulose. In one aspect, the carbohydrate binding module is amino acids 363 to 399 of SEQ ID NO: 2.

Catalase: The term "catalase" means a hydrogen-peroxide:hydrogen-peroxide oxidoreductase (E.C. 1.11.1.6 or E.C. 1.11.1.21) that catalyzes the conversion of two hydrogen peroxides to oxygen and two waters.

Catalase activity can be determined by monitoring the degradation of hydrogen peroxide at 240 nm based on the following reaction:



The reaction is conducted in 50 mM phosphate pH 7 at 25° C. with 10.3 mM substrate (H_2O_2). Absorbance is monitored spectrophotometrically within 16-24 seconds, which should correspond to an absorbance reduction from 0.45 to 0.4. One catalase activity unit can be expressed as one pmole of H_2O_2 degraded per minute at pH 7.0 and 25° C.

Catalytic domain: The term "catalytic domain" means the region of an enzyme containing the catalytic machinery of the enzyme. In one aspect, the catalytic domain is amino acids 20 to 341 of SEQ ID NO: 2.

Cellobiohydrolase: The term "cellobiohydrolase" means a 1,4-beta-D-glucan cellobiohydrolase (E.C. 3.2.1.91 and E.C. 3.2.1.176) that catalyzes the hydrolysis of 1,4-beta-D-glucosidic linkages in cellulose, cellooligosaccharides, or any beta-1,4-linked glucose containing polymer, releasing cellobiose from the reducing end (cellobiohydrolase I) or non-reducing end (cellobiohydrolase II) of the chain (Teed, 1997, *Trends in Biotechnology* 15: 160-167; Teeri et al., 1998, *Biochem. Soc. Trans.* 26: 173-178). Cellobiohydrolase activity can be determined according to the procedures described by Lever et al., 1972, *Anal. Biochem.* 47: 273-279; van Tilbeurgh et al., 1982, *FEBS Letters* 149: 152-156; van Tilbeurgh and Claeysens, 1985, *FEBS Letters* 187: 283-288; and Tomme et al., 1988, *Eur. J. Biochem.* 170: 575-581.

Cellulolytic enzyme or cellulase: The term "cellulolytic enzyme" or "cellulase" means one or more enzymes that hydrolyze a cellulosic material. Such enzymes include endoglucanase(s), cellobiohydrolase(s), beta-glucosidase(s), or combinations thereof. The two basic approaches for measuring cellulolytic enzyme activity include: (1) measuring the total cellulolytic enzyme activity, and (2) measuring the individual cellulolytic enzyme activities (endoglucanases, cellobiohydrolases, and beta-glucosidases) as reviewed in Zhang et al., 2006, *Biotechnology Advances* 24: 452-481. Total cellulolytic enzyme activity can be measured using insoluble substrates, including Whatman No 1 filter paper, microcrystalline cellulose, bacterial cellulose, algal cellulose, cotton, pretreated lignocellulose, etc. The most common total cellulolytic activity assay is the filter paper assay using Whatman No 1 filter paper as the substrate. The assay was established by the International Union of Pure and Applied Chemistry (IUPAC) (Ghose, 1987, *Pure Appl. Chem.* 59: 257-68).

Cellulolytic enzyme activity can be determined by measuring the increase in production/release of sugars during hydrolysis of a cellulosic material by cellulolytic enzyme(s) under the following conditions: 1-50 mg of cellulolytic enzyme protein/g of cellulose in pretreated corn stover (PCS) (or other pretreated cellulosic material) for 3-7 days at a suitable temperature such as 40° C.-80° C., e.g., 40° C., 45° C., 50° C., 55° C., 60° C., 65° C., 70° C., 75° C., or 80° C., and a suitable pH, such as 4-9, e.g., 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, or 9.0, compared to a control hydrolysis without addition of cellulolytic enzyme protein. Typical conditions are 1 ml reactions, washed or unwashed PCS, 5% insoluble solids (dry weight), 50 mM sodium acetate pH

5, 1 mM MnSO₄, 50° C., 55° C., or 60° C., 72 hours, sugar analysis by AMINEX® HPX-87H column chromatography (Bio-Rad Laboratories, Inc.).

Cellulosic material: The term “cellulosic material” means any material containing cellulose. The predominant polysaccharide in the primary cell wall of biomass is cellulose, the second most abundant is hemicellulose, and the third is pectin. The secondary cell wall, produced after the cell has stopped growing, also contains polysaccharides and is strengthened by polymeric lignin covalently cross-linked to hemicellulose. Cellulose is a homopolymer of anhydrocellobiose and thus a linear beta-(1-4)-D-glucan, while hemicelluloses include a variety of compounds, such as xylans, xyloglucans, arabinoxylans, and mannans in complex branched structures with a spectrum of substituents. Although generally polymorphous, cellulose is found in plant tissue primarily as an insoluble crystalline matrix of parallel glucan chains. Hemicelluloses usually hydrogen bond to cellulose, as well as to other hemicelluloses, which help stabilize the cell wall matrix.

Cellulose is generally found, for example, in the stems, leaves, hulls, husks, and cobs of plants or leaves, branches, and wood of trees. The cellulosic material can be, but is not limited to, agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, and wood (including forestry residue) (see, for example, Wiselogle et al., 1995, in *Handbook on Bioethanol* (Charles E. Wyman, editor), pp. 105-118, Taylor & Francis, Washington D.C.; Wyman, 1994, *Bioresource Technology* 50: 3-16; Lynd, 1990, *Applied Biochemistry and Biotechnology* 24/25: 695-719; Mosier et al., 1999, Recent Progress in Bioconversion of Lignocelluloses, in *Advances in Biochemical Engineering/Biotechnology*, T. Scheper, managing editor, Volume 65, pp. 23-40, Springer-Verlag, New York). It is understood herein that the cellulose may be in the form of lignocellulose, a plant cell wall material containing lignin, cellulose, and hemicellulose in a mixed matrix. In one aspect, the cellulosic material is any biomass material. In another aspect, the cellulosic material is lignocellulose, which comprises cellulose, hemicelluloses, and lignin.

In an embodiment, the cellulosic material is agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, or wood (including forestry residue).

In another embodiment, the cellulosic material is arundo, bagasse, bamboo, corn cob, corn fiber, corn stover, miscanthus, rice straw, sugar cane straw, switchgrass, or wheat straw.

In another embodiment, the cellulosic material is aspen, eucalyptus, fir, pine, poplar, spruce, or willow.

In another embodiment, the cellulosic material is algal cellulose, bacterial cellulose, cotton linter, filter paper, microcrystalline cellulose (e.g., AVICEL®), or phosphoric-acid treated cellulose.

In another embodiment, the cellulosic material is an aquatic biomass. As used herein the term “aquatic biomass” means biomass produced in an aquatic environment by a photosynthesis process. The aquatic biomass can be algae, emergent plants, floating-leaf plants, or submerged plants.

The cellulosic material may be used as is or may be subjected to pretreatment, using conventional methods known in the art, as described herein. In a preferred aspect, the cellulosic material is pretreated.

Coding sequence: The term “coding sequence” means a polynucleotide, which directly specifies the amino acid sequence of a polypeptide. The boundaries of the coding

sequence are generally determined by an open reading frame that begins with a start codon, such as ATG, GTG, or TTG, and ends with a stop codon, such as TAA, TAG, or TGA. The coding sequence may be a genomic DNA, cDNA, synthetic DNA, or a combination thereof.

Control sequences: The term “control sequences” means nucleic acid sequences necessary for expression of a polynucleotide encoding a polypeptide of the present invention. Each control sequence may be native (i.e., from the same gene) or heterologous (i.e., from a different gene) to the polynucleotide encoding the polypeptide or native or heterologous to each other. Such control sequences include, but are not limited to, a leader, polyadenylation sequence, propeptide sequence, promoter, signal peptide sequence, and transcription terminator. At a minimum, the control sequences include a promoter, and transcriptional and translational stop signals. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the polynucleotide encoding a polypeptide.

Dissolved Oxygen Saturation Level: The saturation level of oxygen is determined at the standard partial pressure (0.21 atmosphere) of oxygen. The saturation level at the standard partial pressure of oxygen is dependent on the temperature and solute concentrations. In an embodiment where the temperature during hydrolysis is 50° C., the saturation level would typically be in the range of 5-5.5 mg oxygen per kg slurry, depending on the solute concentrations. Hence, a concentration of dissolved oxygen of 0.5 to 10% of the saturation level at 50° C. corresponds to an amount of dissolved oxygen in a range from 0.025 ppm (0.5×5/100) to 0.55 ppm (10×5.5/100), such as, e.g., 0.05 to 0.165 ppm, and a concentration of dissolved oxygen of 10-70% of the saturation level at 50° C. corresponds to an amount of dissolved oxygen in a range from 0.50 ppm (10×5/100) to 3.85 ppm (70×5.5/100), such as, e.g., 1 to 2 ppm. In an embodiment, oxygen is added in an amount in the range of 0.5 to 5 ppm, such as 0.5 to 4.5 ppm, 0.5 to 4 ppm, 0.5 to 3.5 ppm, 0.5 to 3 ppm, 0.5 to 2.5 ppm, or 0.5 to 2 ppm.

Endoglucanase: The term “endoglucanase” means a 4-(1,3;1,4)-beta-D-glucan 4-glucanohydrolase (E.C. 3.2.1.4) that catalyzes endohydrolysis of 1,4-beta-D-glycosidic linkages in cellulose, cellulose derivatives (such as carboxymethyl cellulose and hydroxyethyl cellulose), lichenin, beta-1,4 bonds in mixed beta-1,3-1,4 glucans such as cereal beta-D-glucans or xyloglucans, and other plant material containing cellulosic components.

Endoglucanase activity can be determined by measuring reduction in substrate viscosity or increase in reducing ends determined by a reducing sugar assay (Zhang et al., 2006, *Biotechnology Advances* 24: 452-481). Endoglucanase activity can also be determined using carboxymethyl cellulose (CMC) as substrate according to the procedure of Ghose, 1987, *Pure and Appl. Chem.* 59: 257-268, at pH 5, 40° C.

Expression: The term “expression” includes any step involved in the production of a polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, post-translational modification, and secretion.

Expression vector: The term “expression vector” means a linear or circular DNA molecule that comprises a polynucleotide encoding a polypeptide and is operably linked to control sequences that provide for its expression.

Feruloyl esterase: The term “feruloyl esterase” means a 4-hydroxy-3-methoxycinnamoyl-sugar hydrolase (EC

3.1.1.73) that catalyzes the hydrolysis of 4-hydroxy-3-methoxycinnamoyl (feruloyl) groups from esterified sugar, which is usually arabinose in natural biomass substrates, to produce ferulate (4-hydroxy-3-methoxycinnamate). Feruloyl esterase (FAE) is also known as ferulic acid esterase, hydroxycinnamoyl esterase, FAE-III, cinnamoyl ester hydrolase, FAEA, cinnAE, FAE-I, or FAE-II. Feruloyl esterase activity can be determined using 0.5 mM p-nitrophenylferulate as substrate in 50 mM sodium acetate pH 5.0. One unit of feruloyl esterase equals the amount of enzyme capable of releasing 1 μ mole of p-nitrophenolate anion per minute at pH 5, 25° C.

Fragment: The term “fragment” means a polypeptide or a catalytic domain or carbohydrate binding module having one or more amino acids absent from the amino and/or carboxyl terminus of a mature polypeptide or domain or module; wherein the fragment has xylanase. In one aspect, a fragment having xylanase activity contains at least 320 amino acid residues, at least 340 amino acid residues, or at least 360 amino acid residues of the mature polypeptide of SEQ ID NO: 2.

Fusion polypeptide: The term “fusion polypeptide” is a polypeptide in which one polypeptide is fused at the N-terminus or the C-terminus of a polypeptide of the present invention. A fusion polypeptide is produced by fusing a polynucleotide encoding another polypeptide to a polynucleotide of the present invention. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fusion polypeptide is under control of the same promoter(s) and terminator. Fusion polypeptides may also be constructed using intein technology in which fusion polypeptides are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779). A fusion polypeptide can further comprise a cleavage site between the two polypeptides. Upon secretion of the fusion protein, the site is cleaved releasing the two polypeptides. Examples of cleavage sites include, but are not limited to, the sites disclosed in Martin et al., 2003, *J. Ind. Microbiol. Biotechnol.* 3: 568-576; Svetina et al., 2000, *J. Biotechnol.* 76: 245-251; Rasmussen-Wilson et al., 1997, *Appl. Environ. Microbiol.* 63: 3488-3493; Ward et al., 1995, *Biotechnology* 13: 498-503; and Contreras et al., 1991, *Biotechnology* 9: 378-381; Eaton et al., 1986, *Biochemistry* 25: 505-512; Collins-Racie et al., 1995, *Biotechnology* 13: 982-987; Carter et al., 1989, *Proteins: Structure, Function, and Genetics* 6: 240-248; and Stevens, 2003, *Drug Discovery World* 4: 35-48.

Hemicellulolytic enzyme or hemicellulase: The term “hemicellulolytic enzyme” or “hemicellulase” means one or more enzymes that hydrolyze a hemicellulosic material. See, for example, Shallom and Shoham, 2003, *Current Opinion In Microbiology* 6(3): 219-228). Hemicellulases are key components in the degradation of plant biomass. Examples of hemicellulases include, but are not limited to, an acetylmannan esterase, an acetylxylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. The substrates for these enzymes, hemicelluloses, are a heterogeneous group of branched and linear polysaccharides that are bound via hydrogen bonds to the cellulose microfibrils in the plant cell wall, crosslinking them into a robust network. Hemicelluloses are also covalently attached to lignin, forming together with cellulose a highly complex structure. The variable structure and orga-

nization of hemicelluloses require the concerted action of many enzymes for its complete degradation. The catalytic modules of hemicellulases are either glycoside hydrolases (GHs) that hydrolyze glycosidic bonds, or carbohydrate esterases (CEs), which hydrolyze ester linkages of acetate or ferulic acid side groups. These catalytic modules, based on homology of their primary sequence, can be assigned into GH and CE families. Some families, with an overall similar fold, can be further grouped into clans, marked alphabetically (e.g., GH-A). A most informative and updated classification of these and other carbohydrate active enzymes is available in the Carbohydrate-Active Enzymes (CAZY) database. Hemicellulolytic enzyme activities can be measured according to Ghose and Bisaria, 1987, *Pure & Appl. Chem.* 59: 1739-1752, at a suitable temperature such as 40° C.-80° C., e.g., 40° C., 45° C., 50° C., 55° C., 60° C., 65° C., 70° C., 75° C., or 80° C., and a suitable pH such as 4-9, e.g., 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, or 9.0.

Hemicellulosic material: The term “hemicellulosic material” means any material comprising hemicelluloses. Hemicelluloses include xylan, glucuronoxylan, arabinoxylan, glucomannan, and xyloglucan. These polysaccharides contain many different sugar monomers. Sugar monomers in hemicellulose can include xylose, mannose, galactose, rhamnose, and arabinose. Hemicelluloses contain most of the D-pentose sugars. Xylose is in most cases the sugar monomer present in the largest amount, although in softwoods mannose can be the most abundant sugar. Xylan contains a backbone of beta-(1-4)-linked xylose residues. Xylans of terrestrial plants are heteropolymers possessing a beta-(1-4)-D-xylopyranose backbone, which is branched by short carbohydrate chains. They comprise D-glucuronic acid or its 4-O-methyl ether, L-arabinose, and/or various oligosaccharides, composed of D-xylose, L-arabinose, D- or L-galactose, and D-glucose. Xylan-type polysaccharides can be divided into homoxylans and heteroxylans, which include glucuronoxylans, (arabino)glucuronoxylans, (glucurono)arabinoxylans, arabinoxylans, and complex heteroxylans. See, for example, Ebringerova et al., 2005, *Adv. Polym. Sci.* 186: 1-67. Hemicellulosic material is also known herein as “xylan-containing material”.

Sources for hemicellulosic material are essentially the same as those for cellulosic material described herein.

In the processes of the present invention, any material containing hemicellulose may be used. In a preferred aspect, the hemicellulosic material is lignocellulose.

Host cell: The term “host cell” means any cell type that is susceptible to transformation, transfection, transduction, or the like with a nucleic acid construct or expression vector comprising a polynucleotide of the present invention. The term “host cell” encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication.

Hybrid polypeptide: The term “hybrid polypeptide” means a polypeptide comprising domains from two or more polypeptides, e.g., a binding module from one polypeptide and a catalytic domain from another polypeptide. The domains may be fused at the N-terminus or the C-terminus.

Hybridization: The term “hybridization” means the pairing of substantially complementary strands of nucleic acids, using standard Southern blotting procedures. Hybridization may be performed under medium, medium-high, high or very high stringency conditions. Medium stringency conditions means prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 35% formamide for 12 to 24 hours, followed by washing three times each for 15 minutes

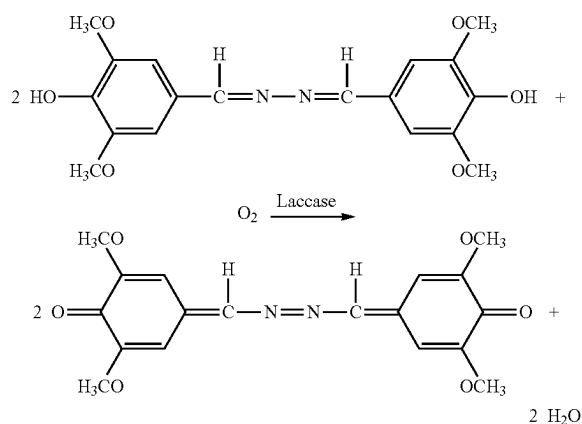
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using 0.2×SSC, 0.2% SDS at 55° C. Medium-high stringency conditions means prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 35% formamide for 12 to 24 hours, followed by washing three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 60° C. High stringency conditions means prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 50% formamide for 12 to 24 hours, followed by washing three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 65° C. Very high stringency conditions means prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 50% formamide for 12 to 24 hours, followed by washing three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 70° C.

Isolated: The term “isolated” means a polypeptide, nucleic acid, cell, or other specified material or component that is separated from at least one other material or component with which it is naturally associated as found in nature, including but not limited to, for example, other proteins, nucleic acids, cells, etc. An isolated polypeptide includes, but is not limited to, a culture broth containing the secreted polypeptide.

Laccase: The term “laccase” means a benzenediol:oxygen oxidoreductase (E.C. 1.10.3.2) that catalyzes the following reaction: 1,2- or 1,4-benzenediol+O₂→1,2- or 1,4-benzo-semiquinone+2H₂O.

Laccase activity can be determined by the oxidation of syringaldazine (4,4'-[azinobis(methanylylidene)]bis(2,6-dimethoxyphenol)) to the corresponding quinone 4,4'-[azobis(methanylylidene)]bis(2,6-dimethoxycyclohexa-2,5-dien-1-one) by laccase. The reaction (shown below) is detected by an increase in absorbance at 530 nm.



The reaction is conducted in 23 mM MES pH 5.5 at 30° C. with 19 μM substrate (syringaldazine) and 1 g/L polyethylene glycol (PEG) 6000. The sample is placed in a spectrophotometer and the change in absorbance is measured at 530 nm every 15 seconds up to 90 seconds. One laccase unit is the amount of enzyme that catalyzes the conversion of 1 μmole syringaldazine per minute under the specified analytical conditions.

Mature polypeptide: The term “mature polypeptide” means a polypeptide in its mature form following removal of a signal peptide by N-terminal processing. In one aspect, the mature polypeptide is amino acids 20 to 399 of SEQ ID NO:

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2 based on the SignalP 3.0 program (Bendtsen et al., 2004, *J. Mol. Biol.* 340: 783-795) or SignalP 4.0 program (Petersen et al., 2011, *Nature Methods* 8: 785-786) that predicts amino acids 1 to 19 of SEQ ID NO: 2 are a signal peptide. In another aspect, the mature polypeptide is SEQ ID NO: 3.

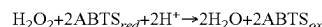
Mature polypeptide coding sequence: The term “mature polypeptide coding sequence” means a polynucleotide that encodes a mature polypeptide having xylanase activity. In one aspect, the mature polypeptide coding sequence is nucleotides 108 to 1415 of SEQ ID NO: 1 based on the SignalP 3.0 program (Bendtsen et al., 2004, *supra*) or SignalP 4.0 program (Petersen et al., 2011, *supra*) that predicts nucleotides 1 to 107 of SEQ ID NO: 1 encode a signal peptide.

Nucleic acid construct: The term “nucleic acid construct” means a nucleic acid molecule, either single- or double-stranded, which is isolated from a naturally occurring gene or is modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature or which is synthetic, which comprises one or more control sequences.

Operably linked: The term “operably linked” means a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

Peroxidase: The term “peroxidase” means an enzyme that converts a peroxide, e.g., hydrogen peroxide, to a less oxidative species, e.g., water. It is understood herein that a peroxidase encompasses a peroxide-decomposing enzyme. The term “peroxide-decomposing enzyme” is defined herein as a donor:peroxide oxidoreductase (E.C. number 1.11.1.x, wherein x=1-3, 5, 7-19, or 21) that catalyzes the reaction reduced substrate(2e⁻)+ROOR'→oxidized substrate+ROH+R'OH; such as horseradish peroxidase that catalyzes the reaction phenol+H₂O₂→quinone+H₂O, and catalase that catalyzes the reaction H₂O₂+H₂O₂→O₂+2H₂O. In addition to hydrogen peroxide, other peroxides may also be decomposed by these enzymes.

Peroxidase activity can be determined by measuring the oxidation of 2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid (ABTS) by a peroxidase in the presence of hydrogen peroxide as shown below. The reaction product ABTS_{ox} forms a blue-green color which can be quantified at 418 nm.



The reaction is conducted in 0.1 M phosphate pH 7 at 30° C. with 1.67 mM substrate (ABTS), 1.5 g/L TRITON® X-405, 0.88 mM hydrogen peroxide, and approximately 0.040 units enzyme per ml. The sample is placed in a spectrophotometer and the change in absorbance is measured at 418 nm from 15 seconds up to 60 seconds. One peroxidase unit can be expressed as the amount of enzyme required to catalyze the conversion of 1 μmole of hydrogen peroxide per minute under the specified analytical conditions.

Pretreated cellulosic or hemicellulosic material: The term “pretreated cellulosic or hemicellulosic material” means a cellulosic or hemicellulosic material derived from biomass by treatment with heat and dilute sulfuric acid, alkaline pretreatment, neutral pretreatment, or any pretreatment known in the art.

Pretreated corn stover: The term “Pretreated Corn Stover” or “PCS” means a cellulosic material derived from corn stover by treatment with heat and dilute sulfuric acid, alkaline pretreatment, neutral pretreatment, or any pretreatment known in the art.

Sequence identity: The relatedness between two amino acid sequences or between two nucleotide sequences is described by the parameter "sequence identity".

For purposes of the present invention, the sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16: 276-277), preferably version 5.0.0 or later. The parameters used are a gap open penalty of 10, a gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled "longest identity" (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Residues} \times 100)}{(\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})}$$

For purposes of the present invention, the sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, supra) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, supra), preferably version 5.0.0 or later. The parameters used are a gap open penalty of 10, a gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled "longest identity" (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Deoxyribonucleotides} \times 100)}{(\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})}$$

Subsequence: The term "subsequence" means a polynucleotide having one or more nucleotides absent from the 5' and/or 3' end of a mature polypeptide coding sequence; wherein the subsequence encodes a fragment having xylanase activity. In one aspect, a subsequence contains at least 960 nucleotides, at least 1020 nucleotides, or at least 1080 nucleotides of the mature polypeptide coding sequence of SEQ ID NO: 1.

Variant: The term "variant" means a polypeptide having xylanase activity comprising an alteration, i.e., a substitution, insertion, and/or deletion (e.g., truncation), at one or more (e.g., several) positions. A substitution means replacement of the amino acid occupying a position with a different amino acid; a deletion means removal of the amino acid occupying a position; and an insertion means adding an amino acid adjacent to and immediately following the amino acid occupying a position.

Xylan-containing material: The term "xylan-containing material" means any material comprising a plant cell wall polysaccharide containing a backbone of beta-(1-4)-linked xylose residues. Xylans of terrestrial plants are heteropolymers possessing a beta-(1-4)-D-xylopyranose backbone, which is branched by short carbohydrate chains. They comprise D-glucuronic acid or its 4-O-methyl ether, L-arabinose, and/or various oligosaccharides, composed of D-xylose, L-arabinose, D- or L-galactose, and D-glucose. Xylan-type polysaccharides can be divided into homoxylans and heteroxylans, which include glucuronoxylans, (arabino)glucuronoxylans, (glucurono)arabinoxylans, arabinoxylans, and complex heteroxylans. See, for example, Ebringerova et al., 2005, *Adv. Polym. Sci.* 186: 1-67.

In the processes of the present invention, any material containing xylan may be used. In a preferred aspect, the xylan-containing material is lignocellulose.

Xylan degrading activity or xylanolytic activity: The term "xylan degrading activity" or "xylanolytic activity" means a biological activity that hydrolyzes xylan-containing material. The two basic approaches for measuring xylanolytic activity include: (1) measuring the total xylanolytic activity, and (2) measuring the individual xylanolytic activities (e.g., endoxylanases, beta-xylosidases, arabinofuranosidases, alpha-glucuronidases, acetylxyylan esterases, feruloyl esterases, and alpha-glucuronyl esterases). Recent progress in assays of xylanolytic enzymes was summarized in several publications including Biely and Puchard, 2006, *Journal of the Science of Food and Agriculture* 86(11): 1636-1647; Spanikova and Biely, 2006, *FEBS Letters* 580(19): 4597-4601; Herrmann et al., 1997, *Biochemical Journal* 321: 375-381.

Total xylan degrading activity can be measured by determining the reducing sugars formed from various types of xylan, including, for example, oat spelt, beechwood, and larchwood xylans, or by photometric determination of dyed xylan fragments released from various covalently dyed xylans. A common total xylanolytic activity assay is based on production of reducing sugars from polymeric 4-O-methyl glucuronoxylan as described in Bailey et al., 1992, *Interlaboratory testing of methods for assay of xylanase activity, Journal of Biotechnology* 23(3): 257-270. Xylanase activity can also be determined with 0.2% AZCL-arabinoxylan as substrate in 0.01% TRITON® X-100 and 200 mM sodium phosphate pH 6 at 37° C. One unit of xylanase activity is defined as 1.0 µmole of azurine produced per minute at 37° C., pH 6 from 0.2% AZCL-arabinoxylan as substrate in 200 mM sodium phosphate pH 6.

Xylan degrading activity can be determined by measuring the increase in hydrolysis of birchwood xylan (Sigma Chemical Co., Inc.) by xylan-degrading enzyme(s) under the following typical conditions: 1 ml reactions, 5 mg/ml substrate (total solids), 5 mg of xylanolytic protein/g of substrate, 50 mM sodium acetate pH 5, 50° C., 24 hours, sugar analysis using p-hydroxybenzoic acid hydrazide (PHBAH) assay as described by Lever, 1972, *Anal. Biochem.* 47: 273-279.

Xylanase: The term "xylanase" means a 1,4-beta-D-xylan-xylohydrolase (E.C. 3.2.1.8) that catalyzes the endohydrolysis of 1,4-beta-D-xylosidic linkages in xylans. Xylanase activity can be determined with 0.44% azo-arabinoxylan as substrate in 50 mM succinate-NaOH pH 6.2, 75 mM NaCl, 0.01% TRITON® X-100 at 50° C. One unit of xylanase activity is defined as 1.0 µmole of Remazol Brilliant Blue R produced per minute in this buffer at 50° C. Xylanase activity can also be determined according to Example 7 or 11 herein.

In one aspect, the polypeptides of the present invention have at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the xylanase activity of the mature polypeptide of SEQ ID NO: 2.

DETAILED DESCRIPTION OF THE INVENTION

Polypeptides Having Xylanase Activity

In an embodiment, the present invention relates to isolated polypeptides having a sequence identity to the mature polypeptide of SEQ ID NO: 2 of at least 99% or 100%, which have xylanase activity. In one aspect, the polypeptides

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differ by up to 10 amino acids, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, from the mature polypeptide of SEQ ID NO: 2.

A polypeptide of the present invention preferably comprises or consists of the amino acid sequence of SEQ ID NO: 2, or an allelic variant thereof; or is a fragment thereof having xylanase activity. In one aspect, the polypeptide comprises or consists of the mature polypeptide of SEQ ID NO: 2. In another aspect, the polypeptide comprises or consists of amino acids 20 to 399 of SEQ ID NO: 2. In another aspect, the polypeptide comprises or consists of

SEQ ID NO: 3. In another embodiment, the present invention relates to isolated polypeptides having xylanase activity encoded by polynucleotides that hybridize under medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the mature polypeptide coding sequence of SEQ ID NO: 1, or the full-length complement thereof (Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, N.Y.).

The polynucleotide of SEQ ID NO: 1, or a subsequence thereof, as well as the polypeptide of SEQ ID NO: 2, or a fragment thereof, may be used to design nucleic acid probes to identify and clone DNA encoding polypeptides having xylanase activity from strains of different genera or species according to methods well known in the art. In particular, such probes can be used for hybridization with the genomic DNA or cDNA of a cell of interest, following standard Southern blotting procedures, in order to identify and isolate the corresponding gene therein. Such probes can be considerably shorter than the entire sequence, but should be at least 15, e.g., at least 25, at least 35, or at least 70 nucleotides in length. Preferably, the nucleic acid probe is at least 100 nucleotides in length, e.g., at least 200 nucleotides, at least 300 nucleotides, at least 400 nucleotides, at least 500 nucleotides, at least 600 nucleotides, at least 700 nucleotides, at least 800 nucleotides, or at least 900 nucleotides in length. Both DNA and RNA probes can be used. The probes are typically labeled for detecting the corresponding gene (for example, with ^{32}P , ^3H , ^{35}S , biotin, or avidin). Such probes are encompassed by the present invention.

A genomic DNA or cDNA library prepared from such other strains may be screened for DNA that hybridizes with the probes described above and encodes a polypeptide having xylanase activity. Genomic or other DNA from such other strains may be separated by agarose or polyacrylamide gel electrophoresis, or other separation techniques. DNA from the libraries or the separated DNA may be transferred to and immobilized on nitrocellulose or another suitable carrier material. In order to identify a clone or DNA that hybridizes with SEQ ID NO: 1, or a subsequence thereof, the carrier material is used in a Southern blot.

For purposes of the present invention, hybridization indicates that the polynucleotides hybridize to a labeled nucleic acid probe corresponding to (i) SEQ ID NO: 1; (ii) the mature polypeptide coding sequence of SEQ ID NO: 1; (iii) the full-length complement thereof; or (iv) a subsequence thereof; under very low to very high stringency conditions. Molecules to which the nucleic acid probe hybridizes under these conditions can be detected using, for example, X-ray film or any other detection means known in the art.

In one aspect, the nucleic acid probe is nucleotides 108 to 1415 of SEQ ID NO: 1 or a subsequence thereof. In another aspect, the nucleic acid probe is a polynucleotide that encodes the polypeptide of SEQ ID NO: 2; the mature polypeptide thereof; or a fragment thereof. In another aspect, the nucleic acid probe is SEQ ID NO: 1.

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In another embodiment, the present invention relates to isolated polypeptides having xylanase activity encoded by polynucleotides having a sequence identity to the mature polypeptide coding sequence of SEQ ID NO: 1 of at least 99% or 100%.

In another embodiment, the present invention relates to variants of the mature polypeptide of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions. In one aspect, the number of amino acid substitutions, deletions and/or insertions introduced into the mature polypeptide of SEQ ID NO: 2 is up to 10, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10. The amino acid changes may be of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of 1-30 amino acids; small amino- or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, *The Proteins*, Academic Press, New York. Common substitutions are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the polypeptide, alter the substrate specificity, change the pH optimum, and the like.

Essential amino acids in a polypeptide can be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant molecules are tested for xylanase activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site of the enzyme or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312; Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identity of essential amino acids can also be inferred from an alignment with a related polypeptide.

Single or multiple amino acid substitutions, deletions, and/or insertions can be made and tested using known methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO 95/17413; or WO 95/22625. Other methods that can be used include error-prone PCR, phage

display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO 92/06204), and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7: 127).

Mutagenesis/shuffling methods can be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active polypeptides can be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.

In some embodiments, the polypeptide is a fragment having xylanase activity containing at least 320 amino acid residues, at least 340 amino acid residues, or at least 360 amino acid residues of the mature polypeptide of SEQ ID NO: 2.

The polypeptide may be a hybrid polypeptide or a fusion polypeptide.

Sources of Polypeptides Having Xylanase Activity

A polypeptide having xylanase activity of the present invention may be obtained from microorganisms of any genus. For purposes of the present invention, the term "obtained from" as used herein in connection with a given source shall mean that the polypeptide encoded by a polynucleotide is produced by the source or by a strain in which the polynucleotide from the source has been inserted. In one aspect, the polypeptide obtained from a given source is secreted extracellularly.

In one aspect, the polypeptide is a *Neosartorya* polypeptide. In another aspect, the polypeptide is a *Neosartorya aurata*, *Neosartorya botucatensi*, *Neosartorya coreana*, *Neosartorya dlicata*, *Neosartorya fennelliae*, *Neosartorya fischeri*, *Neosartorya fumigata*, *Neosartorya glabra*, *Neosartorya hiratsukae*, *Neosartorya laciniola*, *Neosartorya massa*, *Neosartorya multiplicata*, *Neosartorya pseudofischeri*, *Neosartorya primulina*, *Neosartorya quadricincta*, *Neosartorya spathulate*, *Neosartorya spinosa*, *Neosartorya stramenia*, *Neosartorya sublevispora*, *Neosartorya tatenoi*, or *Neosartorya udagawae* polypeptide.

In another aspect, the polypeptide is a *Neosartorya aurata* polypeptide. In another aspect, the polypeptide is a *Neosartorya botucatensi* polypeptide. In another aspect, the polypeptide is a *Neosartorya coreana* polypeptide. In another aspect, the polypeptide is a *Neosartorya dlicata* polypeptide. In another aspect, the polypeptide is a *Neosartorya fennelliae* polypeptide. In another aspect, the polypeptide is a *Neosartorya fischeri* polypeptide. In another aspect, the polypeptide is a *Neosartorya fumigata* polypeptide. In another aspect, the polypeptide is a *Neosartorya glabra* polypeptide. In another aspect, the polypeptide is a *Neosartorya hiratsukae* polypeptide. In another aspect, the polypeptide is a *Neosartorya laciniola* polypeptide. In another aspect, the polypeptide is a *Neosartorya massa* polypeptide. In another aspect, the polypeptide is a *Neosartorya multiplicata* polypeptide. In another aspect, the polypeptide is a *Neosartorya pseudofischeri* polypeptide. In another aspect, the polypeptide is a *Neosartorya primulina* polypeptide. In another aspect, the polypeptide is a *Neosartorya quadricincta* polypeptide. In another aspect, the polypeptide is a *Neosartorya spathulate* polypeptide. In another aspect, the polypeptide is a *Neosartorya spinosa* polypeptide. In another aspect, the polypeptide is a *Neosartorya stramenia* polypeptide. In another aspect, the polypeptide is a *Neosartorya sublevispora* polypeptide. In another aspect, the poly-

peptide is a *Neosartorya tatenoi* polypeptide. In another aspect, the polypeptide is a *Neosartorya udagawae* polypeptide.

It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs, regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

The polypeptides may be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. A polynucleotide encoding the polypeptide may then be obtained by similarly screening a genomic DNA or cDNA library of another microorganism or mixed DNA sample. Once a polynucleotide encoding a polypeptide has been detected with the probe(s), the polynucleotide can be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, *supra*).

Catalytic Domains

In one embodiment, the present invention also relates to catalytic domains having a sequence identity to amino acids 20 to 341 of SEQ ID NO: 2 of at least 99% or 100%. In one aspect, the catalytic domains comprise amino acid sequences that differ by up to 10 amino acids, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, from amino acids 20 to 341 of SEQ ID NO: 2.

The catalytic domain preferably comprises or consists of amino acids 20 to 341 of SEQ ID NO: 2 or an allelic variant thereof; or is a fragment thereof having xylanase activity.

In another embodiment, the present invention also relates to catalytic domains encoded by polynucleotides that hybridize under medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions (as defined above) with the nucleotides 108 to 1241 of SEQ ID NO: 1 or the full-length complement thereof (Sambrook et al., 1989, *supra*).

In another embodiment, the present invention also relates to catalytic domains encoded by polynucleotides having a sequence identity to nucleotides 108 to 1241 of SEQ ID NO: 1 of at least 99% or 100%.

The polynucleotide encoding the catalytic domain preferably comprises or consists of nucleotides 108 to 1241 of SEQ ID NO: 1.

In another embodiment, the present invention also relates to catalytic domain variants of amino acids 20 to 341 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions. In one aspect, the number of amino acid substitutions, deletions and/or insertions introduced into the sequence of amino acids 20 to 341 of SEQ ID NO: 2 is up to 10, e.g., 1, 2, 3, 4, 5, 6, 8, 9, or 10.

In another aspect, a polypeptide comprising a catalytic domain of the present invention may further comprise a carbohydrate binding module.

Carbohydrate Binding Modules

In one embodiment, the present invention also relates to carbohydrate binding modules having a sequence identity to amino acids 363 to 399 of SEQ ID NO: 2 of at least 99% or 100%. In one aspect, the carbohydrate binding modules comprise amino acid sequences that differ by up to 10 amino acids, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, from amino acids 363 to 399 of SEQ ID NO: 2.

The carbohydrate binding module preferably comprises or consists of amino acids 363 to 399 of SEQ ID NO: 2 or an allelic variant thereof; or is a fragment thereof having carbohydrate binding activity.

In another embodiment, the present invention also relates to carbohydrate binding modules encoded by polynucleotides that hybridize under medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions (as defined above) with the nucleotides 1305 to 1415 of SEQ ID NO: 1, or the full-length complement thereof (Sambrook et al., 1989, supra).

In another embodiment, the present invention also relates to carbohydrate binding modules encoded by polynucleotides having a sequence identity to nucleotides 1305 to 1415 of SEQ ID NO: 1 of at least 99% or 100%.

The polynucleotide encoding the carbohydrate binding module preferably comprises or consists of nucleotides 1305 to 1415 of SEQ ID NO: 1.

In another embodiment, the present invention also relates to carbohydrate binding module variants of amino acids 363 to 399 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions. In one aspect, the number of amino acid substitutions, deletions and/or insertions introduced into the sequence of amino acids 363 to 399 of SEQ ID NO: 2 is up to 10, e.g., 1, 2, 3, 4, 5, 6, 8, 9, or 10.

The catalytic domain operably linked to the carbohydrate binding module may be from a hydrolase, isomerase, ligase, lyase, oxidoreductase, or transferase, e.g., an aminopeptidase, amylase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, alpha-galactosidase, beta-galactosidase, glucoamylase, alpha-glucosidase, beta-glucosidase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, xylanase, or beta-xylosidase. The polynucleotide encoding the catalytic domain may be obtained from any prokaryotic, eukaryotic, or other source.

The polypeptides may further comprise a linker between the catalytic domain and the carbohydrate binding module. Polynucleotides

The present invention also relates to isolated polynucleotides encoding a polypeptide, a catalytic domain, or carbohydrate binding module of the present invention, as described herein.

The techniques used to isolate or clone a polynucleotide are known in the art and include isolation from genomic DNA or cDNA, or a combination thereof. The cloning of the polynucleotides from genomic DNA can be effected, e.g., by using the well-known polymerase chain reaction (PCR) or antibody screening of expression libraries to detect cloned DNA fragments with shared structural features. See, e.g., Innis et al., 1990, *PCR: A Guide to Methods and Application*, Academic Press, New York. Other nucleic acid amplification procedures such as ligase chain reaction (LCR), ligation activated transcription (LAT) and polynucleotide-

based amplification (NASBA) may be used. The polynucleotides may be cloned from a strain of *Neosartorya*, or a related organism and thus, for example, may be an allelic or species variant of the polypeptide encoding region of the polynucleotide.

Modification of a polynucleotide encoding a polypeptide of the present invention may be necessary for synthesizing polypeptides substantially similar to the polypeptide. The term "substantially similar" to the polypeptide refers to non-naturally occurring forms of the polypeptide. These polypeptides may differ in some engineered way from the polypeptide isolated from its native source, e.g., variants that differ in specific activity, thermostability, pH optimum, or the like. The variants may be constructed on the basis of the polynucleotide presented as the mature polypeptide coding sequence of SEQ ID NO: 1, e.g., a subsequence thereof, and/or by introduction of nucleotide substitutions that do not result in a change in the amino acid sequence of the polypeptide, but which correspond to the codon usage of the host organism intended for production of the enzyme, or by introduction of nucleotide substitutions that may give rise to a different amino acid sequence. For a general description of nucleotide substitution, see, e.g., Ford et al., 1991, *Protein Expression and Purification* 2: 95-107.

Nucleic Acid Constructs

The present invention also relates to nucleic acid constructs comprising a polynucleotide of the present invention, wherein the polynucleotide is operably linked to one or more control sequences that direct the expression of the coding sequence in a suitable host cell under conditions compatible with the control sequences.

The polynucleotide may be manipulated in a variety of ways to provide for expression of the polypeptide. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

The control sequence may be a promoter, a polynucleotide that is recognized by a host cell for expression of a polynucleotide encoding a polypeptide of the present invention. The promoter contains transcriptional control sequences that mediate the expression of the polypeptide. The promoter may be any polynucleotide that shows transcriptional activity in the host cell including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the host cell.

Examples of suitable promoters for directing transcription of the polynucleotides of the present invention in a bacterial host cell are the promoters obtained from the *Bacillus amyloliquefaciens* alpha-amylase gene (amyQ), *Bacillus licheniformis* alpha-amylase gene (amyL), *Bacillus licheniformis* penicillinase gene (penP), *Bacillus stearothermophilus* maltogenic amylase gene (amyM), *Bacillus subtilis* levansucrase gene (sacB), *Bacillus subtilis* xylA and xylB genes, *Bacillus thuringiensis* cryIIIA gene (Agaisse and Lereclus, 1994, *Molecular Microbiology* 13: 97-107), *E. coli* lac operon, *E. coli* trc promoter (Egon et al., 1988, *Gene* 69: 301-315), *Streptomyces coelicolor* agarase gene (dagA), and prokaryotic beta-lactamase gene (Villa-Kamaroff et al., 1978, *Proc. Natl. Acad. Sci. USA* 75: 3727-3731), as well as the tac promoter (DeBoer et al., 1983, *Proc. Natl. Acad. Sci. USA* 80: 21-25). Further promoters are described in "Useful proteins from recombinant bacteria" in Gilbert et al., 1980, *Scientific American* 242: 74-94; and in Sambrook et al., 1989, supra. Examples of tandem promoters are disclosed in WO 99/43835.

Examples of suitable promoters for directing transcription of the polynucleotides of the present invention in a filamentous fungal host cell are promoters obtained from the genes for *Aspergillus nidulans* acetamidase, *Aspergillus niger* neutral alpha-amylase, *Aspergillus niger* acid stable alpha-amylase, *Aspergillus niger* or *Aspergillus awamori* glucoamylase (glaA), *Aspergillus oryzae* TAKA amylase, *Aspergillus oryzae* alkaline protease, *Aspergillus oryzae* triose phosphate isomerase, *Fusarium oxysporum* trypsin-like protease (WO 96/00787), *Fusarium venenatum* amyloglucosidase (WO 00/56900), *Fusarium venenatum* Daria (WO 00/56900), *Fusarium venenatum* Quinn (WO 00/56900), *Rhizomucor miehei* lipase, *Rhizomucor miehei* aspartic proteinase, *Trichoderma reesei* beta-glucosidase, *Trichoderma reesei* cellobiohydrolase I, *Trichoderma reesei* cellobiohydrolase II, *Trichoderma reesei* endoglucanase I, *Trichoderma reesei* endoglucanase II, *Trichoderma reesei* endoglucanase III, *Trichoderma reesei* endoglucanase V, *Trichoderma reesei* xylanase I, *Trichoderma reesei* xylanase II, *Trichoderma reesei* xylanase III, *Trichoderma reesei* beta-xylosidase, and *Trichoderma reesei* translation elongation factor, as well as the NA2-tpi promoter (a modified promoter from an *Aspergillus* neutral alpha-amylase gene in which the untranslated leader has been replaced by an untranslated leader from an *Aspergillus* triose phosphate isomerase gene; non-limiting examples include modified promoters from an *Aspergillus niger* neutral alpha-amylase gene in which the untranslated leader has been replaced by an untranslated leader from an *Aspergillus nidulans* or *Aspergillus oryzae* triose phosphate isomerase gene); and mutant, truncated, and hybrid promoters thereof. Other promoters are described in U.S. Pat. No. 6,011,147.

In a yeast host, useful promoters are obtained from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* galactokinase (GAL1), *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH1, ADH2/GAP), *Saccharomyces cerevisiae* triose phosphate isomerase (TPI), *Saccharomyces cerevisiae* metallothionein (CUP1), and *Saccharomyces cerevisiae* 3-phosphoglycerate kinase. Other useful promoters for yeast host cells are described by Romanos et al., 1992, *Yeast* 8: 423-488.

The control sequence may also be a transcription terminator, which is recognized by a host cell to terminate transcription. The terminator is operably linked to the 3'-terminus of the polynucleotide encoding the polypeptide. Any terminator that is functional in the host cell may be used in the present invention.

Preferred terminators for bacterial host cells are obtained from the genes for *Bacillus clausii* alkaline protease (aprH), *Bacillus licheniformis* alpha-amylase (amyL), and *Escherichia coli* ribosomal RNA (rrnB).

Preferred terminators for filamentous fungal host cells are obtained from the genes for *Aspergillus nidulans* acetamidase, *Aspergillus nidulans* anthranilate synthase, *Aspergillus niger* glucoamylase, *Aspergillus niger* alpha-glucosidase, *Aspergillus oryzae* TAKA amylase, *Fusarium oxysporum* trypsin-like protease, *Trichoderma reesei* beta-glucosidase, *Trichoderma reesei* cellobiohydrolase I, *Trichoderma reesei* cellobiohydrolase II, *Trichoderma reesei* endoglucanase I, *Trichoderma reesei* endoglucanase II, *Trichoderma reesei* endoglucanase III, *Trichoderma reesei* endoglucanase V, *Trichoderma reesei* xylanase I, *Trichoderma reesei* xylanase II, *Trichoderma reesei* xylanase III, *Trichoderma reesei* beta-xylosidase, and *Trichoderma reesei* translation elongation factor.

Preferred terminators for yeast host cells are obtained from the genes for *Saccharomyces cerevisiae* enolase, *Saccharomyces cerevisiae* cytochrome C (CYC1), and *Saccharomyces cerevisiae* glyceraldehyde-3-phosphate dehydrogenase. Other useful terminators for yeast host cells are described by Romanos et al., 1992, *supra*.

The control sequence may also be an mRNA stabilizer region downstream of a promoter and upstream of the coding sequence of a gene which increases expression of the gene.

Examples of suitable mRNA stabilizer regions are obtained from a *Bacillus thuringiensis* cryIIIA gene (WO 94/25612) and a *Bacillus subtilis* SP82 gene (Hue et al., 1995, *Journal of Bacteriology* 177: 3465-3471).

The control sequence may also be a leader, a nontranslated region of an mRNA that is important for translation by the host cell. The leader is operably linked to the 5'-terminus of the polynucleotide encoding the polypeptide. Any leader that is functional in the host cell may be used.

Preferred leaders for filamentous fungal host cells are obtained from the genes for *Aspergillus oryzae* TAKA amylase and *Aspergillus nidulans* triose phosphate isomerase.

Suitable leaders for yeast host cells are obtained from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* 3-phosphoglycerate kinase, *Saccharomyces cerevisiae* alpha-factor, and *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP).

The control sequence may also be a polyadenylation sequence, a sequence operably linked to the 3'-terminus of the polynucleotide and, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA. Any polyadenylation sequence that is functional in the host cell may be used.

Preferred polyadenylation sequences for filamentous fungal host cells are obtained from the genes for *Aspergillus nidulans* anthranilate synthase, *Aspergillus niger* glucoamylase, *Aspergillus niger* alpha-glucosidase, *Aspergillus oryzae* TAKA amylase, and *Fusarium oxysporum* trypsin-like protease.

Useful polyadenylation sequences for yeast host cells are described by Guo and Sherman, 1995, *Mol. Cellular Biol.* 15: 5983-5990.

The control sequence may also be a signal peptide coding region that encodes a signal peptide linked to the N-terminus of a polypeptide and directs the polypeptide into the cell's secretory pathway. The 5'-end of the coding sequence of the polynucleotide may inherently contain a signal peptide coding sequence naturally linked in translation reading frame with the segment of the coding sequence that encodes the polypeptide. Alternatively, the 5'-end of the coding sequence may contain a signal peptide coding sequence that is heterologous to the coding sequence. A heterologous signal peptide coding sequence may be required where the coding sequence does not naturally contain a signal peptide coding sequence. Alternatively, a heterologous signal peptide coding sequence may simply replace the natural signal peptide coding sequence to enhance secretion of the polypeptide. However, any signal peptide coding sequence that directs the expressed polypeptide into the secretory pathway of a host cell may be used.

Effective signal peptide coding sequences for bacterial host cells are the signal peptide coding sequences obtained from the genes for *Bacillus* NCIB 11837 maltogenic amylase, *Bacillus licheniformis* subtilisin, *Bacillus licheniformis* beta-lactamase, *Bacillus stearothermophilus* alpha-amylase,

Bacillus stearothermophilus neutral proteases (nprT, nprS, nprM), and *Bacillus subtilis* prsA. Further signal peptides are described by Simonen and Palva, 1993, *Microbiological Reviews* 57: 109-137.

Effective signal peptide coding sequences for filamentous fungal host cells are the signal peptide coding sequences obtained from the genes for *Aspergillus niger* neutral amylase, *Aspergillus niger* glucoamylase, *Aspergillus oryzae* TAKA amylase, *Humicola insolens* cellulase, *Humicola insolens* endoglucanase V, *Humicola lanuginosa* lipase, and *Rhizomucor miehei* aspartic proteinase.

Useful signal peptides for yeast host cells are obtained from the genes for *Saccharomyces cerevisiae* alpha-factor and *Saccharomyces cerevisiae* invertase. Other useful signal peptide coding sequences are described by Romanos et al., 1992, *supra*.

The control sequence may also be a propeptide coding sequence that encodes a propeptide positioned at the N-terminus of a polypeptide. The resultant polypeptide is known as a proenzyme or propolypeptide (or a zymogen in some cases). A propolypeptide is generally inactive and can be converted to an active polypeptide by catalytic or autocatalytic cleavage of the propeptide from the propolypeptide. The propeptide coding sequence may be obtained from the genes for *Bacillus subtilis* alkaline protease (aprE), *Bacillus subtilis* neutral protease (nprT), *Myceliophthora thermophila* lactase (WO 95/33836), *Rhizomucor miehei* aspartic proteinase, and *Saccharomyces cerevisiae* alpha-factor.

Where both signal peptide and propeptide sequences are present, the propeptide sequence is positioned next to the N-terminus of a polypeptide and the signal peptide sequence is positioned next to the N-terminus of the propeptide sequence.

It may also be desirable to add regulatory sequences that regulate expression of the polypeptide relative to the growth of the host cell. Examples of regulatory sequences are those that cause expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound. Regulatory sequences in prokaryotic systems include the lac, tac, and trp operator systems. In yeast, the ADH2 system or GAL1 system may be used. In filamentous fungi, the *Aspergillus niger* glucoamylase promoter, *Aspergillus oryzae* TAKA alpha-amylase promoter, and *Aspergillus oryzae* glucoamylase promoter, *Trichoderma reesei* cellobiohydrolase I promoter, and *Trichoderma reesei* cellobiohydrolase II promoter may be used. Other examples of regulatory sequences are those that allow for gene amplification. In eukaryotic systems, these regulatory sequences include the dihydrofolate reductase gene that is amplified in the presence of methotrexate, and the metallothionein genes that are amplified with heavy metals. In these cases, the polynucleotide encoding the polypeptide would be operably linked to the regulatory sequence.

Expression Vectors

The present invention also relates to recombinant expression vectors comprising a polynucleotide of the present invention, a promoter, and transcriptional and translational stop signals. The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more convenient restriction sites to allow for insertion or substitution of the polynucleotide encoding the polypeptide at such sites. Alternatively, the polynucleotide may be expressed by inserting the polynucleotide or a nucleic acid construct comprising the polynucleotide into an appropriate vector for expression. In creating the expression vector, the coding sequence is

located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

The vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the host cell, or a transposon, may be used.

The vector preferably contains one or more selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like.

Examples of bacterial selectable markers are *Bacillus licheniformis* or *Bacillus subtilis* dal genes, or markers that confer antibiotic resistance such as ampicillin, chloramphenicol, kanamycin, neomycin, spectinomycin, or tetracycline resistance. Suitable markers for yeast host cells include, but are not limited to, ADE2, HIS3, LEU2, LYS2, MET3, TRP1, and URA3. Selectable markers for use in a filamentous fungal host cell include, but are not limited to, adeA (phosphoribosylaminoimidazole-succinocarboxamide synthase), adeB (phosphoribosylaminoimidazole synthase), amdS (acetamidase), argB (ornithine carbamoyltransferase), bar (phosphinothricin acetyltransferase), hph (hygromycin phosphotransferase), niaD (nitrate reductase), pyrG (orotidine-5'-phosphate decarboxylase), sC (sulfate adenylyltransferase), and trpC (anthranilate synthase), as well as equivalents thereof. Preferred for use in an *Aspergillus* cell are *Aspergillus nidulans* or *Aspergillus oryzae* amdS and pyrG genes and a *Streptomyces hygroscopicus* bar gene. Preferred for use in a *Trichoderma* cell are adeA, adeB, amdS, hph, and pyrG genes.

The selectable marker may be a dual selectable marker system as described in WO 2010/039889. In one aspect, the dual selectable marker is a hph-tk dual selectable marker system.

The vector preferably contains an element(s) that permits integration of the vector into the host cell's genome or autonomous replication of the vector in the cell independent of the genome.

For integration into the host cell genome, the vector may rely on the polynucleotide's sequence encoding the polypeptide or any other element of the vector for integration into the genome by homologous or non-homologous recombination. Alternatively, the vector may contain additional polynucleotides for directing integration by homologous recombination into the genome of the host cell at a precise location(s) in the chromosome(s). To increase the likelihood of integration at a precise location, the integrational elements should contain a sufficient number of nucleic acids,

such as 100 to 10,000 base pairs, 400 to 10,000 base pairs, and 800 to 10,000 base pairs, which have a high degree of sequence identity to the corresponding target sequence to enhance the probability of homologous recombination. The integrational elements may be any sequence that is homologous with the target sequence in the genome of the host cell. Furthermore, the integrational elements may be non-encoding or encoding polynucleotides. On the other hand, the vector may be integrated into the genome of the host cell by non-homologous recombination.

For autonomous replication, the vector may further comprise an origin of replication enabling the vector to replicate autonomously in the host cell in question. The origin of replication may be any plasmid replicator mediating autonomous replication that functions in a cell. The term "origin of replication" or "plasmid replicator" means a polynucleotide that enables a plasmid or vector to replicate in vivo.

Examples of bacterial origins of replication are the origins of replication of plasmids pBR322, pUC19, pACYC177, and pACYC184 permitting replication in *E. coli*, and pUB110, pE194, pTA1060, and pAM β 1 permitting replication in *Bacillus*.

Examples of origins of replication for use in a yeast host cell are the 2 micron origin of replication, ARS1, ARS4, the combination of ARS1 and CEN3, and the combination of ARS4 and CEN6.

Examples of origins of replication useful in a filamentous fungal cell are AMA1 and ANS1 (Gems et al., 1991, *Gene* 98: 61-67; Cullen et al., 1987, *Nucleic Acids Res.* 15: 9163-9175; WO 00/24883). Isolation of the AMA1 gene and construction of plasmids or vectors comprising the gene can be accomplished according to the methods disclosed in WO 00/24883.

More than one copy of a polynucleotide of the present invention may be inserted into a host cell to increase production of a polypeptide. An increase in the copy number of the polynucleotide can be obtained by integrating at least one additional copy of the sequence into the host cell genome or by including an amplifiable selectable marker gene with the polynucleotide where cells containing amplified copies of the selectable marker gene, and thereby additional copies of the polynucleotide, can be selected for by cultivating the cells in the presence of the appropriate selectable agent.

The procedures used to ligate the elements described above to construct the recombinant expression vectors of the present invention are well known to one skilled in the art (see, e.g., Sambrook et al., 1989, *supra*).

Host Cells

The present invention also relates to recombinant host cells, comprising a polynucleotide of the present invention operably linked to one or more control sequences that direct the production of a polypeptide of the present invention. A construct or vector comprising a polynucleotide is introduced into a host cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier. The term "host cell" encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication. The choice of a host cell will to a large extent depend upon the gene encoding the polypeptide and its source.

The host cell may be any cell useful in the recombinant production of a polypeptide of the present invention, e.g., a prokaryote or a eukaryote.

The prokaryotic host cell may be any Gram-positive or Gram-negative bacterium. Gram-positive bacteria include,

but are not limited to, *Bacillus*, *Clostridium*, *Enterococcus*, *Geobacillus*, *Lactobacillus*, *Lactococcus*, *Oceanobacillus*, *Staphylococcus*, *Streptococcus*, and *Streptomyces*. Gram-negative bacteria include, but are not limited to, *Campylobacter*, *E. coli*, *Flavobacterium*, *Fusobacterium*, *Helicobacter*, *Ilyobacter*, *Neisseria*, *Pseudomonas*, *Salmonella*, and *Ureaplasma*.

The bacterial host cell may be any *Bacillus* cell including, but not limited to, *Bacillus alkalophilus*, *Bacillus amyloliquefaciens*, *Bacillus brevis*, *Bacillus circulans*, *Bacillus clausii*, *Bacillus coagulans*, *Bacillus firmus*, *Bacillus lautus*, *Bacillus lentus*, *Bacillus licheniformis*, *Bacillus megaterium*, *Bacillus pumilus*, *Bacillus stearothermophilus*, *Bacillus subtilis*, and *Bacillus thuringiensis* cells.

The bacterial host cell may also be any *Streptococcus* cell including, but not limited to, *Streptococcus equisimilis*, *Streptococcus pyogenes*, *Streptococcus uberis*, and *Streptococcus equi* subsp. *Zooepidemicus* cells.

The bacterial host cell may also be any *Streptomyces* cell including, but not limited to, *Streptomyces achromogenes*, *Streptomyces avermitilis*, *Streptomyces coelicolor*, *Streptomyces griseus*, and *Streptomyces lividans* cells.

The introduction of DNA into a *Bacillus* cell may be effected by protoplast transformation (see, e.g., Chang and Cohen, 1979, *Mol. Gen. Genet.* 168: 111-115), competent cell transformation (see, e.g., Young and Spizizen, 1961, *J. Bacteriol.* 81: 823-829, or Dubnau and Davidoff-Abelson, 1971, *J. Mol. Biol.* 56: 209-221), electroporation (see, e.g., Shigekawa and Dower, 1988, *Biotechniques* 6: 742-751), or conjugation (see, e.g., Koehler and Thorne, 1987, *J. Bacteriol.* 169: 5271-5278). The introduction of DNA into an *E. coli* cell may be effected by protoplast transformation (see, e.g., Hanahan, 1983, *J. Mol. Biol.* 166: 557-580) or electroporation (see, e.g., Dower et al., 1988, *Nucleic Acids Res.* 16: 6127-6145). The introduction of DNA into a *Streptomyces* cell may be effected by protoplast transformation, electroporation (see, e.g., Gong et al., 2004, *Folia Microbiol. (Praha)* 49: 399-405), conjugation (see, e.g., Mazodier et al., 1989, *J. Bacteriol.* 171: 3583-3585), or transduction (see, e.g., Burke et al., 2001, *Proc. Natl. Acad. Sci. USA* 98: 6289-6294). The introduction of DNA into a *Pseudomonas* cell may be effected by electroporation (see, e.g., Choi et al., 2006, *J. Microbiol. Methods* 64: 391-397) or conjugation (see, e.g., Pinedo and Smets, 2005, *Appl. Environ. Microbiol.* 71: 51-57). The introduction of DNA into a *Streptococcus* cell may be effected by natural competence (see, e.g., Perry and Kuramitsu, 1981, *Infect. Immun.* 32: 1295-1297), protoplast transformation (see, e.g., Catt and Jollick, 1991, *Microbios* 68: 189-207), electroporation (see, e.g., Buckley et al., 1999, *Appl. Environ. Microbiol.* 65: 3800-3804), or conjugation (see, e.g., Clewell, 1981, *Microbiol. Rev.* 45: 409-436). However, any method known in the art for introducing DNA into a host cell can be used.

The host cell may also be a eukaryote, such as a mammalian, insect, plant, or fungal cell.

The host cell may be a fungal cell. "Fungi" as used herein includes the phyla Ascomycota, Basidiomycota, Chytridiomycota, and Zygomycota as well as the Oomycota and all mitosporic fungi (as defined by Hawksworth et al., *In, Ainsworth and Bisby's Dictionary of The Fungi*, 8th edition, 1995, CAB International, University Press, Cambridge, UK).

The fungal host cell may be a yeast cell. "Yeast" as used herein includes ascosporogenous yeast (Endomycetales), basidiosporogenous yeast, and yeast belonging to the Fungi Imperfecti (Blastomycetes). Since the classification of yeast may change in the future, for the purposes of this invention,

yeast shall be defined as described in Biology and Activities of Yeast (Skinner, Passmore, and Davenport, editors, *Soc. App. Bacteriol. Symposium Series* No. 9, 1980).

The yeast host cell may be a *Candida*, *Hansenula*, *Kluyveromyces*, *Pichia*, *Saccharomyces*, *Schizosaccharomyces*, or *Yarrowia* cell, such as a *Kluyveromyces lactis*, *Saccharomyces carlsbergensis*, *Saccharomyces cerevisiae*, *Saccharomyces diastaticus*, *Saccharomyces douglasii*, *Saccharomyces kluyveri*, *Saccharomyces norbensis*, *Saccharomyces oviformis*, or *Yarrowia lipolytica* cell.

The fungal host cell may be a filamentous fungal cell. "Filamentous fungi" include all filamentous forms of the subdivision Eumycota and Oomycota (as defined by Hawksworth et al., 1995, supra). The filamentous fungi are generally characterized by a mycelial wall composed of chitin, cellulose, glucan, chitosan, mannan, and other complex polysaccharides. Vegetative growth is by hyphal elongation and carbon catabolism is obligately aerobic. In contrast, vegetative growth by yeasts such as *Saccharomyces cerevisiae* is by budding of a unicellular thallus and carbon catabolism may be fermentative.

The filamentous fungal host cell may be an *Acremonium*, *Aspergillus*, *Aureobasidium*, *Bjerkandera*, *Ceriporiopsis*, *Chrysosporium*, *Coprinus*, *Coriolus*, *Cryptococcus*, *Filobasidium*, *Fusarium*, *Humicola*, *Magnaporthe*, *Mucor*, *Myceliophthora*, *Neocallimastix*, *Neurospora*, *Paecilomyces*, *Penicillium*, *Phanerochaete*, *Phlebia*, *Piromyces*, *Pleurotus*, *Schizophyllum*, *Talaromyces*, *Thermoascus*, *Thielavia*, *Tolyposcladium*, *Trametes*, or *Trichoderma* cell.

For example, the filamentous fungal host cell may be an *Aspergillus awamori*, *Aspergillus foetidus*, *Aspergillus fumigatus*, *Aspergillus japonicus*, *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus oryzae*, *Bjerkandera adusta*, *Ceriporiopsis aneirina*, *Ceriporiopsis caregiea*, *Ceriporiopsis gilvescens*, *Ceriporiopsis pannocinta*, *Ceriporiopsis rivulosa*, *Ceriporiopsis subrufa*, *Ceriporiopsis subvermisporea*, *Chrysosporium inops*, *Chrysosporium keratinophilum*, *Chrysosporium lucknowense*, *Chrysosporium merdarium*, *Chrysosporium pannicola*, *Chrysosporium queenslandicum*, *Chrysosporium tropicum*, *Chrysosporium zonatum*, *Coprinus cinereus*, *Coriolus hirsutus*, *Fusarium bactridioides*, *Fusarium cerealis*, *Fusarium crookwellense*, *Fusarium culmorum*, *Fusarium graminearum*, *Fusarium gramineum*, *Fusarium heterosporum*, *Fusarium negundi*, *Fusarium oxysporum*, *Fusarium reticulatum*, *Fusarium roseum*, *Fusarium sambucinum*, *Fusarium sarcochroum*, *Fusarium sporotrichioides*, *Fusarium sulphureum*, *Fusarium torulosum*, *Fusarium trichothecioides*, *Fusarium venenatum*, *Humicola insolens*, *Humicola lanuginosa*, *Mucor miehei*, *Myceliophthora thermophila*, *Neurospora crassa*, *Penicillium purpurogenum*, *Phanerochaete chrysosporium*, *Phlebia radiata*, *Pleurotus eryngii*, *Talaromyces emersonii*, *Thielavia terrestris*, *Trametes villosa*, *Trametes versicolor*, *Trichoderma harzianum*, *Trichoderma koningii*, *Trichoderma longibrachiatum*, *Trichoderma reesei*, or *Trichoderma viride* cell.

Fungal cells may be transformed by a process involving protoplast formation, transformation of the protoplasts, and regeneration of the cell wall in a manner known per se. Suitable procedures for transformation of *Aspergillus* and *Trichoderma* host cells are described in EP 238023, Yelton et al., 1984, *Proc. Natl. Acad. Sci. USA* 81: 1470-1474, and Christensen et al., 1988, *Bio/Technology* 6: 1419-1422. Suitable methods for transforming *Fusarium* species are described by Malardier et al., 1989, *Gene* 78: 147-156, and WO 96/00787. Yeast may be transformed using the procedures described by Becker and Guarente, In Abelson, J. N.

and Simon, M. I., editors, *Guide to Yeast Genetics and Molecular Biology, Methods in Enzymology*, Volume 194, pp 182-187, Academic Press, Inc., New York; Ito et al., 1983, *J. Bacteriol.* 153: 163; and Hinnen et al., 1978, *Proc. Natl. Acad. Sci. USA* 75: 1920.

Methods of Production

The present invention also relates to methods of producing a polypeptide of the present invention, comprising (a) cultivating a wild-type cell under conditions conducive for production of the polypeptide; and optionally (b) recovering the polypeptide. In one aspect, the cell is a *Neosartorya* cell. In another aspect, the cell is a *Neosartorya massa* cell. In another aspect, the cell is *Neosartorya massa* CBS 117265.

The present invention also relates to methods of producing a polypeptide of the present invention, comprising (a) cultivating a recombinant host cell of the present invention under conditions conducive for production of the polypeptide; and optionally (b) recovering the polypeptide.

The host cells are cultivated in a nutrient medium suitable for production of the polypeptide using methods known in the art. For example, the cells may be cultivated by shake flask cultivation, or small-scale or large-scale fermentation (including continuous, batch, fed-batch, or solid-state fermentations) in laboratory or industrial fermentors in a suitable medium and under conditions allowing the polypeptide to be expressed and/or isolated. The cultivation takes place in a suitable nutrient medium comprising carbon and nitrogen sources and inorganic salts, using procedures known in the art. Suitable media are available from commercial suppliers or may be prepared according to published compositions (e.g., in catalogues of the American Type Culture Collection). If the polypeptide is secreted into the nutrient medium, the polypeptide can be recovered directly from the medium. If the polypeptide is not secreted, it can be recovered from cell lysates.

The polypeptide may be detected using methods known in the art that are specific for the polypeptides. These detection methods include, but are not limited to, use of specific antibodies, formation of an enzyme product, or disappearance of an enzyme substrate. For example, an enzyme assay may be used to determine the activity of the polypeptide.

The polypeptide may be recovered using methods known in the art. For example, the polypeptide may be recovered from the fermentation medium by conventional procedures including, but not limited to, collection, centrifugation, filtration, extraction, spray-drying, evaporation, or precipitation. In one aspect, a whole fermentation broth comprising the polypeptide is recovered.

The polypeptide may be purified by a variety of procedures known in the art including, but not limited to, chromatography (e.g., ion exchange, affinity, hydrophobic, chromatofocusing, and size exclusion), electrophoretic procedures (e.g., preparative isoelectric focusing), differential solubility (e.g., ammonium sulfate precipitation), SDS-PAGE, or extraction (see, e.g., *Protein Purification*, Janson and Ryden, editors, VCH Publishers, New York, 1989) to obtain substantially pure polypeptides.

In an alternative aspect, the polypeptide is not recovered, but rather a host cell of the present invention expressing the polypeptide is used as a source of the polypeptide.

Plants

The present invention also relates to isolated plants, e.g., a transgenic plant, plant part, or plant cell, comprising a polynucleotide of the present invention so as to express and produce a polypeptide or domain or module in recoverable quantities. The polypeptide or domain or module may be recovered from the plant or plant part. Alternatively, the

plant or plant part containing the polypeptide or domain or module may be used as such for improving the quality of a food or feed, e.g., improving nutritional value, palatability, and rheological properties, or to destroy an antinutritive factor.

The transgenic plant can be dicotyledonous (a dicot) or monocotyledonous (a monocot). Examples of monocot plants are grasses, such as meadow grass (blue grass, *Poa*), forage grass such as *Festuca*, *Lolium*, temperate grass, such as *Agrostis*, and cereals, e.g., wheat, oats, rye, barley, rice, sorghum, and maize (corn).

Examples of dicot plants are tobacco, legumes, such as lupins, potato, sugar beet, pea, bean and soybean, and cruciferous plants (family Brassicaceae), such as cauliflower, rape seed, and the closely related model organism *Arabidopsis thaliana*.

Examples of plant parts are stem, callus, leaves, root, fruits, seeds, and tubers as well as the individual tissues comprising these parts, e.g., epidermis, mesophyll, parenchyme, vascular tissues, meristems. Specific plant cell compartments, such as chloroplasts, apoplasts, mitochondria, vacuoles, peroxisomes and cytoplasm are also considered to be a plant part. Furthermore, any plant cell, whatever the tissue origin, is considered to be a plant part. Likewise, plant parts such as specific tissues and cells isolated to facilitate the utilization of the invention are also considered plant parts, e.g., embryos, endosperms, aleurone and seed coats.

Also included within the scope of the present invention are the progeny of such plants, plant parts, and plant cells.

The transgenic plant or plant cell expressing the polypeptide or domain or module may be constructed in accordance with methods known in the art. In short, the plant or plant cell is constructed by incorporating one or more expression constructs encoding the polypeptide or domain or module into the plant host genome or chloroplast genome and propagating the resulting modified plant or plant cell into a transgenic plant or plant cell. In an embodiment, a plant cell does not belong to plant varieties.

The expression construct is conveniently a nucleic acid construct that comprises a polynucleotide encoding a polypeptide or domain or module, wherein the polynucleotide is operably linked with appropriate regulatory sequences required for expression of the polynucleotide in the plant or plant part of choice. Furthermore, the expression construct may comprise a selectable marker useful for identifying plant cells into which the expression construct has been integrated and DNA sequences necessary for introduction of the construct into the plant in question (the latter depends on the DNA introduction method to be used).

The choice of regulatory sequences, such as promoter and terminator sequences and optionally signal or transit sequences, is determined, for example, on the basis of when, where, and how the polypeptide or domain or module is desired to be expressed (Sticklen, 2008, *Nature Reviews* 9: 433-443). For instance, the expression of the gene encoding a polypeptide or domain or module may be constitutive or inducible, or may be developmental, stage or tissue specific, and the gene product may be targeted to a specific tissue or plant part such as seeds or leaves. Regulatory sequences are, for example, described by Tague et al., 1988, *Plant Physiology* 86: 506.

For constitutive expression, the 35S-CaMV, the maize ubiquitin 1, or the rice actin 1 promoter may be used (Frank et al., 1980, *Cell* 21: 285-294; Christensen et al., 1992, *Plant Mol. Biol.* 18: 675-689; Zhang et al., 1991, *Plant Cell* 3: 1155-1165). Organ-specific promoters may be, for example, a promoter from storage sink tissues such as seeds, potato

tubers, and fruits (Edwards and Coruzzi, 1990, *Ann. Rev. Genet.* 24: 275-303), or from metabolic sink tissues such as meristems (Ito et al., 1994, *Plant Mol. Biol.* 24: 863-878), a seed specific promoter such as the glutelin, prolamin, globulin, or albumin promoter from rice (Wu et al., 1998, *Plant Cell Physiol.* 39: 885-889), a *Vicia faba* promoter from the legumin B4 and the unknown seed protein gene from *Vicia faba* (Conrad et al., 1998, *J. Plant Physiol.* 152: 708-711), a promoter from a seed oil body protein (Chen et al., 1998, *Plant Cell Physiol.* 39: 935-941), the storage protein napA promoter from *Brassica napus*, or any other seed specific promoter known in the art, e.g., as described in WO 91/14772. Furthermore, the promoter may be a leaf specific promoter such as the rbcS promoter from rice or tomato (Kiyozuka et al., 1993, *Plant Physiol.* 102: 991-1000), the *chlorella* virus adenine methyltransferase gene promoter (Mitra and Higgins, 1994, *Plant Mol. Biol.* 26: 85-93), the aldP gene promoter from rice (Kagaya et al., 1995, *Mol. Gen. Genet.* 248: 668-674), or a wound inducible promoter such as the potato pin2 promoter (Xu et al., 1993, *Plant Mol. Biol.* 22: 573-588). Likewise, the promoter may be induced by abiotic treatments such as temperature, drought, or alterations in salinity or induced by exogenously applied substances that activate the promoter, e.g., ethanol, oestrogens, plant hormones such as ethylene, abscisic acid, and gibberellic acid, and heavy metals.

A promoter enhancer element may also be used to achieve higher expression of a polypeptide or domain or module in the plant. For instance, the promoter enhancer element may be an intron that is placed between the promoter and the polynucleotide encoding a polypeptide or domain or module. For instance, Xu et al., 1993, supra, disclose the use of the first intron of the rice actin 1 gene to enhance expression.

The selectable marker gene and any other parts of the expression construct may be chosen from those available in the art.

The nucleic acid construct is incorporated into the plant genome according to conventional techniques known in the art, including *Agrobacterium*-mediated transformation, virus-mediated transformation, microinjection, particle bombardment, biolistic transformation, and electroporation (Gasser et al., 1990, *Science* 244: 1293; Potrykus, 1990, *Bio/Technology* 8: 535; Shimamoto et al., 1989, *Nature* 338: 274).

Agrobacterium tumefaciens-mediated gene transfer is a method for generating transgenic dicots (for a review, see Hooykas and Schilperoort, 1992, *Plant Mol. Biol.* 19: 15-38) and for transforming monocots, although other transformation methods may be used for these plants. A method for generating transgenic monocots is particle bombardment (microscopic gold or tungsten particles coated with the transforming DNA) of embryonic calli or developing embryos (Christou, 1992, *Plant J.* 2: 275-281; Shimamoto, 1994, *Curr. Opin. Biotechnol.* 5: 158-162; Vasil et al., 1992, *Bio/Technology* 10: 667-674). An alternative method for transformation of monocots is based on protoplast transformation as described by Omirulleh et al., 1993, *Plant Mol. Biol.* 21: 415-428. Additional transformation methods include those described in U.S. Pat. Nos. 6,395,966 and 7,151,204 (both of which are herein incorporated by reference in their entirety).

Following transformation, the transformants having incorporated the expression construct are selected and regenerated into whole plants according to methods well known in the art. Often the transformation procedure is designed for the selective elimination of selection genes either during regeneration or in the following generations by

using, for example, co-transformation with two separate T-DNA constructs or site-specific excision of the selection gene by a specific recombinase.

In addition to direct transformation of a particular plant genotype with a construct of the present invention, transgenic plants may be made by crossing a plant having the construct to a second plant lacking the construct. For example, a construct encoding a polypeptide or domain or module can be introduced into a particular plant variety by crossing, without the need for ever directly transforming a plant of that given variety. Therefore, the present invention encompasses not only a plant directly regenerated from cells which have been transformed in accordance with the present invention, but also the progeny of such plants. As used herein, progeny may refer to the offspring of any generation of a parent plant prepared in accordance with the present invention. Such progeny may include a DNA construct prepared in accordance with the present invention. Crossing results in the introduction of a transgene into a plant line by cross pollinating a starting line with a donor plant line. Non-limiting examples of such steps are described in U.S. Pat. No. 7,151,204.

Plants may be generated through a process of backcross conversion. For example, plants include plants referred to as a backcross converted genotype, line, inbred, or hybrid.

Genetic markers may be used to assist in the introgression of one or more transgenes of the invention from one genetic background into another. Marker assisted selection offers advantages relative to conventional breeding in that it can be used to avoid errors caused by phenotypic variations. Further, genetic markers may provide data regarding the relative degree of elite germplasm in the individual progeny of a particular cross. For example, when a plant with a desired trait which otherwise has a non-agronomically desirable genetic background is crossed to an elite parent, genetic markers may be used to select progeny which not only possess the trait of interest, but also have a relatively large proportion of the desired germplasm. In this way, the number of generations required to introgress one or more traits into a particular genetic background is minimized.

The present invention also relates to methods of producing a polypeptide of the present invention comprising (a) cultivating a transgenic plant or a plant cell comprising a polynucleotide encoding the polypeptide under conditions conducive for production of the polypeptide; and (b) recovering the polypeptide.

Fermentation Broth Formulations or Cell Compositions

The present invention also relates to a fermentation broth formulation or a cell composition comprising a polypeptide of the present invention. The fermentation broth product further comprises additional ingredients used in the fermentation process, such as, for example, cells (including, the host cells containing the gene encoding the polypeptide of the present invention which are used to produce the polypeptide), cell debris, biomass, fermentation media and/or fermentation products. In some embodiments, the composition is a cell-killed whole broth containing organic acid(s), killed cells and/or cell debris, and culture medium.

The term "fermentation broth" refers to a preparation produced by cellular fermentation that undergoes no or minimal recovery and/or purification. For example, fermentation broths are produced when microbial cultures are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis (e.g., expression of enzymes by host cells) and secretion into cell culture medium. The fermentation broth can contain unfractionated or fractionated contents of the fermentation materials

derived at the end of the fermentation. Typically, the fermentation broth is unfractionated and comprises the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are removed, e.g., by centrifugation. In some embodiments, the fermentation broth contains spent cell culture medium, extracellular enzymes, and viable and/or nonviable microbial cells.

In an embodiment, the fermentation broth formulation and cell compositions comprise a first organic acid component comprising at least one 1-5 carbon organic acid and/or a salt thereof and a second organic acid component comprising at least one 6 or more carbon organic acid and/or a salt thereof. In a specific embodiment, the first organic acid component is acetic acid, formic acid, propionic acid, a salt thereof, or a mixture of two or more of the foregoing and the second organic acid component is benzoic acid, cyclohexanecarboxylic acid, 4-methylvaleric acid, phenylacetic acid, a salt thereof, or a mixture of two or more of the foregoing.

In one aspect, the composition contains an organic acid (s), and optionally further contains killed cells and/or cell debris. In one embodiment, the killed cells and/or cell debris are removed from a cell-killed whole broth to provide a composition that is free of these components.

The fermentation broth formulations or cell compositions may further comprise a preservative and/or anti-microbial (e.g., bacteriostatic) agent, including, but not limited to, sorbitol, sodium chloride, potassium sorbate, and others known in the art.

The fermentation broth formulations or cell compositions may further comprise multiple enzymatic activities, such as one or more enzymes selected from the group consisting of a cellulase, a hemicellulase, an AA9 polypeptide, a cellulose inducible protein (CIP), a catalase, an esterase, an expansin, a laccase, a ligninolytic enzyme, a pectinase, a peroxidase, a protease, and a swollenin. The fermentation broth formulations or cell compositions may also comprise one or more enzymes selected from the group consisting of a hydrolase, an isomerase, a ligase, a lyase, an oxidoreductase, or a transferase, e.g., an alpha-galactosidase, alpha-glucosidase, aminopeptidase, amylase, beta-galactosidase, beta-glucosidase, beta-xylosidase, carbohydrazase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, glucoamylase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, or xylanase.

The cell-killed whole broth or composition may contain the unfractionated contents of the fermentation materials derived at the end of the fermentation. Typically, the cell-killed whole broth or composition contains the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis (e.g., expression of cellulase and/or glucosidase enzyme(s)). In some embodiments, the cell-killed whole broth or composition contains the spent cell culture medium, extracellular enzymes, and killed filamentous fungal cells. In some embodiments, the microbial cells present in the cell-killed whole broth or composition can be permeabilized and/or lysed using methods known in the art.

A whole broth or cell composition as described herein is typically a liquid, but may contain insoluble components, such as killed cells, cell debris, culture media components, and/or insoluble enzyme(s). In some embodiments, insoluble components may be removed to provide a clarified liquid composition.

The whole broth formulations and cell compositions of the present invention may be produced by a method described in WO 90/15861 or WO 2010/096673.

Enzyme Compositions

The present invention also relates to compositions comprising a polypeptide of the present invention. Preferably, the compositions are enriched in such a polypeptide. The term "enriched" indicates that the amount of the xylanase in the composition has been increased, e.g., with an enrichment factor of at least 1.1.

The compositions may comprise a polypeptide of the present invention as the major enzymatic component, e.g., a mono-component composition. Alternatively, the compositions may comprise multiple enzymatic activities, such as one or more enzymes selected from the group consisting of a cellulase, a hemicellulase, an AA9 polypeptide, a cellulose inducible protein (CIP), a catalase, an esterase, an expansin, a laccase, a ligninolytic enzyme, a pectinase, a peroxidase, a protease, and a swollenin. The compositions may also comprise one or more enzymes selected from the group consisting of a hydrolase, an isomerase, a ligase, a lyase, an oxidoreductase, or a transferase, e.g., an alpha-galactosidase, alpha-glucosidase, aminopeptidase, amylase, beta-galactosidase, beta-glucosidase, beta-xylosidase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, glucoamylase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, or xylanase.

The compositions may be prepared in accordance with methods known in the art and may be in the form of a liquid or a dry composition. The compositions may be stabilized in accordance with methods known in the art.

Examples are given below of uses of the compositions of the present invention. The dosage of the composition and other conditions under which the composition is used may be determined on the basis of methods known in the art.

Uses

The present invention is also directed to the following processes for using the polypeptides having xylanase activity, or compositions thereof.

The present invention also relates to processes for degrading a cellulosic or hemicellulosic material, comprising: treating the cellulosic or hemicellulosic material with an enzyme composition comprising a polypeptide having xylanase activity of the present invention. In one aspect, the processes further comprise recovering the degraded cellulosic or hemicellulosic material. Soluble products from the degradation of the cellulosic or hemicellulosic material can be separated from insoluble cellulosic or hemicellulosic material using methods known in the art such as, for example, centrifugation, filtration, or gravity settling.

The present invention also relates to processes of producing a fermentation product, comprising: (a) saccharifying a cellulosic or hemicellulosic material with an enzyme composition comprising a polypeptide having xylanase activity of the present invention; (b) fermenting the saccharified cellulosic or hemicellulosic material with one or more fermenting microorganisms to produce the fermentation product; and (c) recovering the fermentation product from the fermentation.

The present invention also relates to processes of fermenting a cellulosic or hemicellulosic material, comprising: fermenting the cellulosic or hemicellulosic material with one or more fermenting microorganisms, wherein the cellulosic

or hemicellulosic material is saccharified with an enzyme composition comprising a polypeptide having xylanase activity of the present invention. In one aspect, the fermenting of the cellulosic or hemicellulosic material produces a fermentation product. In another aspect, the processes further comprise recovering the fermentation product from the fermentation.

The processes of the present invention can be used to saccharify the cellulosic or hemicellulosic material to fermentable sugars and to convert the fermentable sugars to many useful fermentation products, e.g., fuel (ethanol, n-butanol, isobutanol, biodiesel, jet fuel) and/or platform chemicals (e.g., acids, alcohols, ketones, gases, oils, and the like). The production of a desired fermentation product from the cellulosic or hemicellulosic material typically involves pretreatment, enzymatic hydrolysis (saccharification), and fermentation.

The processing of the cellulosic or hemicellulosic material according to the present invention can be accomplished using methods conventional in the art. Moreover, the processes of the present invention can be implemented using any conventional biomass processing apparatus configured to operate in accordance with the invention.

Hydrolysis (saccharification) and fermentation, separate or simultaneous, include, but are not limited to, separate hydrolysis and fermentation (SHF); simultaneous saccharification and fermentation (SSF); simultaneous saccharification and co-fermentation (SSCF); hybrid hydrolysis and fermentation (HHF); separate hydrolysis and co-fermentation (SHCF); hybrid hydrolysis and co-fermentation (HHCF); and direct microbial conversion (DMC), also sometimes called consolidated bioprocessing (CBP). SHF uses separate process steps to first enzymatically hydrolyze the cellulosic or hemicellulosic material to fermentable sugars, e.g., glucose, cellobiose, and pentose monomers, and then ferment the fermentable sugars to ethanol. In SSF, the enzymatic hydrolysis of the cellulosic or hemicellulosic material and the fermentation of sugars to ethanol are combined in one step (Philippidis, G. P., 1996, *Cellulose bioconversion technology*, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, D.C., 179-212). SSCF involves the co-fermentation of multiple sugars (Sheehan and Himmel, 1999, *Biotechnol. Prog.* 15: 817-827). HHF involves a separate hydrolysis step, and in addition a simultaneous saccharification and hydrolysis step, which can be carried out in the same reactor. The steps in an HHF process can be carried out at different temperatures, i.e., high temperature enzymatic saccharification followed by SSF at a lower temperature that the fermentation strain can tolerate. DMC combines all three processes (enzyme production, hydrolysis, and fermentation) in one or more steps where the same organism is used to produce the enzymes for conversion of the cellulosic or hemicellulosic material to fermentable sugars and to convert the fermentable sugars into a final product (Lynd et al., 2002, *Microbiol. Mol. Biol. Reviews* 66: 506-577). It is understood herein that any method known in the art comprising pretreatment, enzymatic hydrolysis (saccharification), fermentation, or a combination thereof, can be used in the practicing the processes of the present invention.

A conventional apparatus can include a fed-batch stirred reactor, a batch stirred reactor, a continuous flow stirred reactor with ultrafiltration, and/or a continuous plug-flow column reactor (de Castilhos Corazza et al., 2003, *Acta Scientiarum. Technology* 25: 33-38; Gusakov and Sinitsyn, 1985, *Enz. Microb. Technol.* 7: 346-352), an attrition reactor

(Ryu and Lee, 1983, *Biotechnol. Bioeng.* 25: 53-65). Additional reactor types include fluidized bed, upflow blanket, immobilized, and extruder type reactors for hydrolysis and/or fermentation.

Pretreatment. In practicing the processes of the present invention, any pretreatment process known in the art can be used to disrupt plant cell wall components of the cellulosic or hemicellulosic material (Chandra et al., 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 67-93; Galbe and Zacchi, 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 41-65; Hendriks and Zeeman, 2009, *Bioresource Technology* 100: 10-18; Mosier et al., 2005, *Bioresource Technology* 96: 673-686; Taherzadeh and Karimi, 2008, *Int. J. Mol. Sci.* 9: 1621-1651; Yang and Wyman, 2008, *Biofuels Bioproducts and Biorefining-Biofr.* 2: 26-40).

The cellulosic or hemicellulosic material can also be subjected to particle size reduction, sieving, pre-soaking, wetting, washing, and/or conditioning prior to pretreatment using methods known in the art.

Conventional pretreatments include, but are not limited to, steam pretreatment (with or without explosion), dilute acid pretreatment, hot water pretreatment, alkaline pretreatment, lime pretreatment, wet oxidation, wet explosion, ammonia fiber explosion, organosolv pretreatment, and biological pretreatment. Additional pretreatments include ammonia percolation, ultrasound, electroporation, microwave, supercritical CO₂, supercritical H₂O, ozone, ionic liquid, and gamma irradiation pretreatments.

The cellulosic or hemicellulosic material can be pretreated before hydrolysis and/or fermentation. Pretreatment is preferably performed prior to the hydrolysis. Alternatively, the pretreatment can be carried out simultaneously with enzyme hydrolysis to release fermentable sugars, such as glucose, xylose, and/or cellobiose. In most cases the pretreatment step itself results in some conversion of biomass to fermentable sugars (even in absence of enzymes).

Steam Pretreatment. In steam pretreatment, the cellulosic or hemicellulosic material is heated to disrupt the plant cell wall components, including lignin, hemicellulose, and cellulose to make the cellulose and other fractions, e.g., hemicellulose, accessible to enzymes. The cellulosic or hemicellulosic material is passed to or through a reaction vessel where steam is injected to increase the temperature to the required temperature and pressure and is retained therein for the desired reaction time. Steam pretreatment is preferably performed at 140-250° C., e.g., 160-200° C. or 170-190° C., where the optimal temperature range depends on optional addition of a chemical catalyst. Residence time for the steam pretreatment is preferably 1-60 minutes, e.g., 1-30 minutes, 1-20 minutes, 3-12 minutes, or 4-10 minutes, where the optimal residence time depends on the temperature and optional addition of a chemical catalyst. Steam pretreatment allows for relatively high solids loadings, so that the cellulosic or hemicellulosic material is generally only moist during the pretreatment. The steam pretreatment is often combined with an explosive discharge of the material after the pretreatment, which is known as steam explosion, that is, rapid flashing to atmospheric pressure and turbulent flow of the material to increase the accessible surface area by fragmentation (Duff and Murray, 1996, *Bioresource Technology* 855: 1-33; Galbe and Zacchi, 2002, *Appl. Microbiol. Biotechnol.* 59: 618-628; U.S. Patent Application No. 2002/0164730). During steam pretreatment, hemicellulose acetyl groups are cleaved and the resulting acid autocatalyzes partial hydrolysis of the hemicellulose to monosaccharides and oligosaccharides. Lignin is removed to only a limited extent.

Chemical Pretreatment. The term "chemical treatment" refers to any chemical pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin. Such a pretreatment can convert crystalline cellulose to amorphous cellulose. Examples of suitable chemical pretreatment processes include, for example, dilute acid pretreatment, lime pretreatment, wet oxidation, ammonia fiber/freeze expansion (AFEX), ammonia percolation (APR), ionic liquid, and organosolv pretreatments.

A chemical catalyst such as H₂SO₄ or SO₂ (typically 0.3 to 5% w/w) is sometimes added prior to steam pretreatment, which decreases the time and temperature, increases the recovery, and improves enzymatic hydrolysis (Ballesteros et al., 2006, *Appl. Biochem. Biotechnol.* 129-132: 496-508; Varga et al., 2004, *Appl. Biochem. Biotechnol.* 113-116: 509-523; Sassner et al., 2006, *Enzyme Microb. Technol.* 39: 756-762). In dilute acid pretreatment, the cellulosic or hemicellulosic material is mixed with dilute acid, typically H₂SO₄, and water to form a slurry, heated by steam to the desired temperature, and after a residence time flashed to atmospheric pressure. The dilute acid pretreatment can be performed with a number of reactor designs, e.g., plug-flow reactors, counter-current reactors, or continuous counter-current shrinking bed reactors (Duff and Murray, 1996, *Bioresource Technology* 855: 1-33; Schell et al., 2004, *Bioresource Technology* 91: 179-188; Lee et al., 1999, *Adv. Biochem. Eng. Biotechnol.* 65: 93-115).

Several methods of pretreatment under alkaline conditions can also be used. These alkaline pretreatments include, but are not limited to, sodium hydroxide, lime, wet oxidation, ammonia percolation (APR), and ammonia fiber/freeze expansion (AFEX) pretreatment.

Lime pretreatment is performed with calcium oxide or calcium hydroxide at temperatures of 85-150° C. and residence times from 1 hour to several days (Wyman et al., 2005, *Bioresource Technology* 96: 1959-1966; Mosier et al., 2005, *Bioresource Technology* 96: 673-686). WO 2006/110891, WO 2006/110899, WO 2006/110900, and WO 2006/110901 disclose pretreatment methods using ammonia.

Wet oxidation is a thermal pretreatment performed typically at 180-200° C. for 5-15 minutes with addition of an oxidative agent such as hydrogen peroxide or over-pressure of oxygen (Schmidt and Thomsen, 1998, *Bioresource Technology* 64: 139-151; Palonen et al., 2004, *Appl. Biochem. Biotechnol.* 117: 1-17; Varga et al., 2004, *Biotechnol. Bioeng.* 88: 567-574; Martin et al., 2006, *J. Chem. Technol. Biotechnol.* 81: 1669-1677). The pretreatment is performed preferably at 1-40% dry matter, e.g., 2-30% dry matter or 5-20% dry matter, and often the initial pH is increased by the addition of alkali such as sodium carbonate.

A modification of the wet oxidation pretreatment method, known as wet explosion (combination of wet oxidation and steam explosion) can handle dry matter up to 30%. In wet explosion, the oxidizing agent is introduced during pretreatment after a certain residence time. The pretreatment is then ended by flashing to atmospheric pressure (WO 2006/032282).

Ammonia fiber expansion (AFEX) involves treating the cellulosic or hemicellulosic material with liquid or gaseous ammonia at moderate temperatures such as 90-150° C. and high pressure such as 17-20 bar for 5-10 minutes, where the dry matter content can be as high as 60% (Gollapalli et al., 2002, *Appl. Biochem. Biotechnol.* 98: 23-35; Chundawat et al., 2007, *Biotechnol. Bioeng.* 96: 219-231; Alizadeh et al., 2005, *Appl. Biochem. Biotechnol.* 121: 1133-1141; Teymouri et al., 2005, *Bioresource Technology* 96: 2014-2018).

During AFEX pretreatment cellulose and hemicelluloses remain relatively intact. Lignin-carbohydrate complexes are cleaved.

Organosolv pretreatment delignifies the cellulosic or hemicellulosic material by extraction using aqueous ethanol (40-60% ethanol) at 160-200° C. for 30-60 minutes (Pan et al., 2005, *Biotechnol. Bioeng.* 90: 473-481; Pan et al., 2006, *Biotechnol. Bioeng.* 94: 851-861; Kurabi et al., 2005, *Appl. Biochem. Biotechnol.* 121: 219-230). Sulphuric acid is usually added as a catalyst. In organosolv pretreatment, the majority of hemicellulose and lignin is removed.

Other examples of suitable pretreatment methods are described by Schell et al., 2003, *Appl. Biochem. Biotechnol.* 105-108: 69-85, and Mosier et al., 2005, *Bioresource Technology* 96: 673-686, and U.S. Published Application 2002/0164730.

In one aspect, the chemical pretreatment is preferably carried out as a dilute acid treatment, and more preferably as a continuous dilute acid treatment. The acid is typically sulfuric acid, but other acids can also be used, such as acetic acid, citric acid, nitric acid, phosphoric acid, tartaric acid, succinic acid, hydrogen chloride, or mixtures thereof. Mild acid treatment is conducted in the pH range of preferably 1-5, e.g., 1-4 or 1-2.5. In one aspect, the acid concentration is in the range from preferably 0.01 to 10 wt. % acid, e.g., 0.05 to 5 wt. % acid or 0.1 to 2 wt. % acid. The acid is contacted with the cellulosic or hemicellulosic material and held at a temperature in the range of preferably 140-200° C., e.g., 165-190° C., for periods ranging from 1 to 60 minutes.

In another aspect, pretreatment takes place in an aqueous slurry. In preferred aspects, the cellulosic or hemicellulosic material is present during pretreatment in amounts preferably between 10-80 wt. %, e.g., 20-70 wt. % or 30-60 wt. %, such as around 40 wt. %. The pretreated cellulosic or hemicellulosic material can be unwashed or washed using any method known in the art, e.g., washed with water.

Mechanical Pretreatment or Physical Pretreatment: The term "mechanical pretreatment" or "physical pretreatment" refers to any pretreatment that promotes size reduction of particles. For example, such pretreatment can involve various types of grinding or milling (e.g., dry milling, wet milling, or vibratory ball milling).

The cellulosic or hemicellulosic material can be pretreated both physically (mechanically) and chemically. Mechanical or physical pretreatment can be coupled with steaming/steam explosion, hydrothermolysis, dilute or mild acid treatment, high temperature, high pressure treatment, irradiation (e.g., microwave irradiation), or combinations thereof. In one aspect, high pressure means pressure in the range of preferably about 100 to about 400 psi, e.g., about 150 to about 250 psi. In another aspect, high temperature means temperature in the range of about 100 to about 300° C., e.g., about 140 to about 200° C. In a preferred aspect, mechanical or physical pretreatment is performed in a batch-process using a steam gun hydrolyzer system that uses high pressure and high temperature as defined above, e.g., a Sunds Hydrolyzer available from Sunds Defibrator AB, Sweden. The physical and chemical pretreatments can be carried out sequentially or simultaneously, as desired.

Accordingly, in a preferred aspect, the cellulosic or hemicellulosic material is subjected to physical (mechanical) or chemical pretreatment, or any combination thereof, to promote the separation and/or release of cellulose, hemicellulose, and/or lignin.

Biological Pretreatment. The term "biological pretreatment" refers to any biological pretreatment that promotes the separation and/or release of cellulose, hemicellulose,

and/or lignin from the cellulosic or hemicellulosic material. Biological pretreatment techniques can involve applying lignin-solubilizing microorganisms and/or enzymes (see, for example, Hsu, T.-A., 1996, Pretreatment of biomass, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, D.C., 179-212; Ghosh and Singh, 1993, *Adv. Appl. Microbiol.* 39: 295-333; McMillan, J. D., 1994, Pretreating lignocellulosic biomass: a review, in *Enzymatic Conversion of Biomass for Fuels Production*, Himmel, M. E., Baker, J. O., and Overend, R. P., eds., ACS Symposium Series 566, American Chemical Society, Washington, D.C., chapter 15; Gong, C. S., Cao, N. J., Du, J., and Tsao, G. T., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Olsson and Hahn-Hagerdal, 1996, *Enz. Microb. Tech.* 18: 312-331; and Vallander and Eriksson, 1990, *Adv. Biochem. Eng./Biotechnol.* 42: 63-95).

Saccharification. In the hydrolysis step, also known as saccharification, the cellulosic or hemicellulosic material, e.g., pretreated, is hydrolyzed to break down cellulose and/or hemicellulose to fermentable sugars, such as glucose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides. The hydrolysis is performed enzymatically by one or more enzyme compositions in one or more stages. The hydrolysis can be carried out as a batch process or series of batch processes. The hydrolysis can be carried out as a fed batch or continuous process, or series of fed batch or continuous processes, where the cellulosic or hemicellulosic material is fed gradually to, for example, a hydrolysis solution containing an enzyme composition. In an embodiment the saccharification is a continuous saccharification in which a cellulosic material and a cellulolytic enzyme composition are added at different intervals throughout the saccharification and the hydrolysate is removed at different intervals throughout the saccharification. The removal of the hydrolysate may occur prior to, simultaneously with, or after the addition of the cellulosic material and the cellulolytic enzyme composition.

Enzymatic hydrolysis is preferably carried out in a suitable aqueous environment under conditions that can be readily determined by one skilled in the art. In one aspect, hydrolysis is performed under conditions suitable for the activity of the enzymes(s), i.e., optimal for the enzyme(s).

The saccharification is generally performed in stirred-tank reactors or fermentors under controlled pH, temperature, and mixing conditions. Suitable process time, temperature and pH conditions can readily be determined by one skilled in the art. For example, the total saccharification time can last up to 200 hours, but is typically performed for preferably about 4 to about 120 hours, e.g., about 12 to about 96 hours or about 24 to about 72 hours. The temperature is in the range of preferably about 25° C. to about 80° C., e.g., about 30° C. to about 70° C., about 40° C. to about 60° C., or about 50° C. to about 55° C. The pH is in the range of preferably about 3 to about 9, e.g., about 3.5 to about 8, about 4 to about 7, about 4.2 to about 6, or about 4.3 to about 5.5.

The dry solids content is in the range of preferably about 5 to about 50 wt. %, e.g., about 10 to about 40 wt. % or about 20 to about 30 wt. %.

In one aspect, the saccharification is performed in the presence of dissolved oxygen at a concentration of at least 0.5% of the saturation level.

In an embodiment of the invention the dissolved oxygen concentration during saccharification is in the range of at least 0.5% up to 30% of the saturation level, such as at least

1% up to 25%, at least 1% up to 20%, at least 1% up to 15%, at least 1% up to 10%, at least 1% up to 5%, and at least 1% up to 3% of the saturation level. In a preferred embodiment, the dissolved oxygen concentration is maintained at a concentration of at least 0.5% up to 30% of the saturation level, such as at least 1% up to 25%, at least 1% up to 20%, at least 1% up to 15%, at least 1% up to 10%, at least 1% up to 5%, and at least 1% up to 3% of the saturation level during at least 25%, such as at least 50% or at least 75% of the saccharification period. When the enzyme composition comprises an oxidoreductase the dissolved oxygen concentration may be higher up to 70% of the saturation level.

Oxygen is added to the vessel in order to achieve the desired concentration of dissolved oxygen during saccharification. Maintaining the dissolved oxygen level within a desired range can be accomplished by aeration of the vessel, tank or the like by adding compressed air through a diffuser or sparger, or by other known methods of aeration. The aeration rate can be controlled on the basis of feedback from a dissolved oxygen sensor placed in the vessel/tank, or the system can run at a constant rate without feedback control. In the case of a hydrolysis train consisting of a plurality of vessels/tanks connected in series, aeration can be implemented in one or more or all of the vessels/tanks. Oxygen aeration systems are well known in the art. According to the invention any suitable aeration system may be used. Commercial aeration systems are designed by, e.g., Chemineer, Derby, England, and build by, e.g., Paul Mueller Company, MO, USA.

The enzyme compositions can comprise any protein useful in degrading the cellulosic or hemicellulosic material.

In one aspect, the enzyme composition comprises or further comprises one or more proteins selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin. In another aspect, the cellulase is preferably one or more enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase. In another aspect, the hemicellulase is preferably one or more enzymes selected from the group consisting of an acetylmannan esterase, an acetylxylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. In another aspect, the oxidoreductase is preferably one or more enzymes selected from the group consisting of a catalase, a laccase, and a peroxidase.

In another aspect, the enzyme composition comprises one or more cellulolytic enzymes. In another aspect, the enzyme composition comprises or further comprises one or more hemicellulolytic enzymes. In another aspect, the enzyme composition comprises one or more cellulolytic enzymes and one or more hemicellulolytic enzymes. In another aspect, the enzyme composition comprises one or more enzymes selected from the group of cellulolytic enzymes and hemicellulolytic enzymes. In another aspect, the enzyme composition comprises an endoglucanase. In another aspect, the enzyme composition comprises a cellobiohydrolase. In another aspect, the enzyme composition comprises a beta-glucosidase. In another aspect, the enzyme composition comprises an AA9 polypeptide. In another aspect, the enzyme composition comprises an endoglucanase and an AA9 polypeptide. In another aspect, the enzyme composition comprises a cellobiohydrolase and an AA9 polypeptide. In another aspect, the enzyme composition comprises a beta-glucosidase and an AA9 polypeptide. In another aspect,

the enzyme composition comprises an endoglucanase and a cellobiohydrolase. In another aspect, the enzyme composition comprises an endoglucanase I, an endoglucanase II, or a combination of an endoglucanase I and an endoglucanase II, and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II. In another aspect, the enzyme composition comprises an endoglucanase and a beta-glucosidase. In another aspect, the enzyme composition comprises an endoglucanase I, an endoglucanase II, or a combination of an endoglucanase I and an endoglucanase II, and a beta-glucosidase. In another aspect, the enzyme composition comprises a beta-glucosidase and a cellobiohydrolase. In another aspect, the enzyme composition comprises a beta-glucosidase and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II. In another aspect, the enzyme composition comprises an endoglucanase, an AA9 polypeptide, and a cellobiohydrolase. In another aspect, the enzyme composition comprises an endoglucanase I, an endoglucanase II, or a combination of an endoglucanase I and an endoglucanase II, an AA9 polypeptide, and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II. In another aspect, the enzyme composition comprises an endoglucanase, a beta-glucosidase, and an AA9 polypeptide. In another aspect, the enzyme composition comprises a beta-glucosidase, an AA9 polypeptide, and a cellobiohydrolase. In another aspect, the enzyme composition comprises a beta-glucosidase, an AA9 polypeptide, and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II. In another aspect, the enzyme composition comprises an endoglucanase, a beta-glucosidase, and a cellobiohydrolase. In another aspect, the enzyme composition comprises an endoglucanase I, an endoglucanase II, or a combination of an endoglucanase I and an endoglucanase II, a beta-glucosidase, and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II. In another aspect, the enzyme composition comprises an endoglucanase, a cellobiohydrolase, a beta-glucosidase, and an AA9 polypeptide. In another aspect, the enzyme composition comprises an endoglucanase I, an endoglucanase II, or a combination of an endoglucanase I and an endoglucanase II, a beta-glucosidase, an AA9 polypeptide, and a cellobiohydrolase I, a cellobiohydrolase II, or a combination of a cellobiohydrolase I and a cellobiohydrolase II.

In another aspect, the enzyme composition comprises an acetylmannan esterase. In another aspect, the enzyme composition comprises an acetylxylan esterase. In another aspect, the enzyme composition comprises an arabinanase (e.g., alpha-L-arabinanase). In another aspect, the enzyme composition comprises an arabinofuranosidase (e.g., alpha-L-arabinofuranosidase). In another aspect, the enzyme composition comprises a coumaric acid esterase. In another aspect, the enzyme composition comprises a feruloyl esterase. In another aspect, the enzyme composition comprises a galactosidase (e.g., alpha-galactosidase and/or beta-galactosidase). In another aspect, the enzyme composition comprises a glucuronidase (e.g., alpha-D-glucuronidase). In another aspect, the enzyme composition comprises a glucuronoyl esterase. In another aspect, the enzyme composition comprises a mannanase. In another aspect, the enzyme composition comprises a mannosidase (e.g., beta-mannosidase). In another aspect, the enzyme composition comprises a xylanase. In an embodiment, the xylanase is a Family 10 xylanase. In another embodiment, the xylanase is a Family

11 xylanase. In another aspect, the enzyme composition comprises a xylosidase (e.g., beta-xylosidase).

In another aspect, the enzyme composition comprises an esterase. In another aspect, the enzyme composition comprises an expansin. In another aspect, the enzyme composition comprises a ligninolytic enzyme. In an embodiment, the ligninolytic enzyme is a manganese peroxidase. In another embodiment, the ligninolytic enzyme is a lignin peroxidase. In another embodiment, the ligninolytic enzyme is a H_2O_2 -producing enzyme. In another aspect, the enzyme composition comprises a pectinase. In another aspect, the enzyme composition comprises an oxidoreductase. In an embodiment, the oxidoreductase is a catalase. In another embodiment, the oxidoreductase is a laccase. In another embodiment, the oxidoreductase is a peroxidase. In another aspect, the enzyme composition comprises a protease. In another aspect, the enzyme composition comprises a swollenin.

In the processes of the present invention, the enzyme(s) can be added prior to or during saccharification, saccharification and fermentation, or fermentation.

One or more components of the enzyme composition may be native proteins, recombinant proteins, or a combination of native proteins and recombinant proteins. For example, one or more components may be native proteins of a cell, which is used as a host cell to express recombinantly one or more other components of the enzyme composition. It is understood herein that the recombinant proteins may be heterologous (e.g., foreign) and/or native to the host cell. One or more components of the enzyme composition may be produced as monocomponents, which are then combined to form the enzyme composition. The enzyme composition may be a combination of multicomponent and monocomponent protein preparations.

The enzymes used in the processes of the present invention may be in any form suitable for use, such as, for example, a fermentation broth formulation or a cell composition, a cell lysate with or without cellular debris, a semi-purified or purified enzyme preparation, or a host cell as a source of the enzymes. The enzyme composition may be a dry powder or granulate, a non-dusting granulate, a liquid, a stabilized liquid, or a stabilized protected enzyme. Liquid enzyme preparations may, for instance, be stabilized by adding stabilizers such as a sugar, a sugar alcohol or another polyol, and/or lactic acid or another organic acid according to established processes.

The optimum amounts of the enzymes and polypeptides having xylanase activity depend on several factors including, but not limited to, the mixture of cellulolytic enzymes and/or hemicellulolytic enzymes, the cellulosic or hemicellulosic material, the concentration of cellulosic or hemicellulosic material, the pretreatment(s) of the cellulosic or hemicellulosic material, temperature, time, pH, and inclusion of a fermenting organism (e.g., for Simultaneous Saccharification and Fermentation).

In one aspect, an effective amount of cellulolytic or hemicellulolytic enzyme to the cellulosic or hemicellulosic material is about 0.5 to about 50 mg, e.g., about 0.5 to about 40 mg, about 0.5 to about 25 mg, about 0.75 to about 20 mg, about 0.75 to about 15 mg, about 0.5 to about 10 mg, or about 2.5 to about 10 mg per g of the cellulosic or hemicellulosic material.

In another aspect, an effective amount of a polypeptide having xylanase activity to the cellulosic or hemicellulosic material is about 0.01 to about 50.0 mg, e.g., about 0.01 to about 40 mg, about 0.01 to about 30 mg, about 0.01 to about 20 mg, about 0.01 to about 10 mg, about 0.01 to about 5 mg,

about 0.025 to about 1.5 mg, about 0.05 to about 1.25 mg, about 0.075 to about 1.25 mg, about 0.1 to about 1.25 mg, about 0.15 to about 1.25 mg, or about 0.25 to about 1.0 mg per g of the cellulosic or hemicellulosic material.

In another aspect, an effective amount of a polypeptide having xylanase activity to cellulolytic or hemicellulolytic enzyme is about 0.005 to about 1.0 g, e.g., about 0.01 to about 1.0 g, about 0.15 to about 0.75 g, about 0.15 to about 0.5 g, about 0.1 to about 0.5 g, about 0.1 to about 0.25 g, or about 0.05 to about 0.2 g per g of cellulolytic or hemicellulolytic enzyme.

The polypeptides having cellulolytic enzyme activity or hemicellulolytic enzyme activity as well as other proteins/polypeptides useful in the degradation of the cellulosic or hemicellulosic material, e.g., AA9 polypeptides can be derived or obtained from any suitable origin, including, archaeal, bacterial, fungal, yeast, plant, or animal origin. The term "obtained" also means herein that the enzyme may have been produced recombinantly in a host organism employing methods described herein, wherein the recombinantly produced enzyme is either native or foreign to the host organism or has a modified amino acid sequence, e.g., having one or more amino acids that are deleted, inserted and/or substituted, i.e., a recombinantly produced enzyme that is a mutant and/or a fragment of a native amino acid sequence or an enzyme produced by nucleic acid shuffling processes known in the art. Encompassed within the meaning of a native enzyme are natural variants and within the meaning of a foreign enzyme are variants obtained by, e.g., site-directed mutagenesis or shuffling.

Each polypeptide may be a bacterial polypeptide. For example, each polypeptide may be a Gram-positive bacterial polypeptide having enzyme activity, or a Gram-negative bacterial polypeptide having enzyme activity.

Each polypeptide may also be a fungal polypeptide, e.g., a yeast polypeptide or a filamentous fungal polypeptide.

Chemically modified or protein engineered mutants of polypeptides may also be used.

One or more components of the enzyme composition may be a recombinant component, i.e., produced by cloning of a DNA sequence encoding the single component and subsequent cell transformed with the DNA sequence and expressed in a host (see, for example, WO 91/17243 and WO 91/17244). The host can be a heterologous host (enzyme is foreign to host), but the host may under certain conditions also be a homologous host (enzyme is native to host). Monocomponent cellulolytic proteins may also be prepared by purifying such a protein from a fermentation broth.

In one aspect, the one or more cellulolytic enzymes comprise a commercial cellulolytic enzyme preparation. Examples of commercial cellulolytic enzyme preparations suitable for use in the present invention include, for example, CELLIC® CTec (Novozymes A/S), CELLIC® CTec2 (Novozymes A/S), CELLIC® CTec3 (Novozymes A/S), CELLUCLAST™ (Novozymes A/S), NOVOZYME™ 188 (Novozymes A/S), SPEZYME™ CP (Genencor Int.), ACCELERASE™ TRIO (DuPont), FILTRASE® NL (DSM); METHAPLUS® S/L 100 (DSM), ROHAMENT™ 7069 W (Rohm GmbH), or ALTERNAFUEL® CMAX3™ (Dyadic International, Inc.). The cellulolytic enzyme preparation is added in an amount effective from about 0.001 to about 5.0 wt. % of solids, e.g., about 0.025 to about 4.0 wt. % of solids or about 0.005 to about 2.0 wt. % of solids.

Examples of bacterial endoglucanases that can be used in the processes of the present invention, include, but are not limited to, *Acidothermus cellulolyticus* endoglucanase (WO

91/05039; WO 93/15186; U.S. Pat. No. 5,275,944; WO 96/02551; U.S. Pat. No. 5,536,655; WO 00/70031; WO 2005/093050), *Erwinia carotovora* endoglucanase (Saari-lahti et al., 1990, Gene 90: 9-14), *Thermobifida fusca* endoglucanase III (WO 2005/093050), and *Thermobifida fusca* endoglucanase V (WO 2005/093050).

Examples of fungal endoglucanases that can be used in the present invention, include, but are not limited to, *Trichoderma reesei* endoglucanase I (Penttilä et al., 1986, Gene 45: 253-263, *Trichoderma reesei* Cel7B endoglucanase I (GenBank:M15665), *Trichoderma reesei* endoglucanase II (Saloheimo et al., 1988, Gene 63:11-22), *Trichoderma reesei* Cel5A endoglucanase II (GenBank:M19373), *Trichoderma reesei* endoglucanase III (Okada et al., 1988, Appl. Environ. Microbiol. 64: 555-563, GenBank:AB003694), *Trichoderma reesei* endoglucanase V (Saloheimo et al., 1994, Molecular Microbiology 13: 219-228, GenBank:Z33381), *Aspergillus aculeatus* endoglucanase (Ooi et al., 1990, Nucleic Acids Research 18: 5884), *Aspergillus kawachii* endoglucanase (Sakamoto et al., 1995, Current Genetics 27: 435-439), *Fusarium oxysporum* endoglucanase (GenBank:L29381), *Humicola grisea* var. *thermoidea* endoglucanase (GenBank:AB003107), *Melanocarpus albomyces* endoglucanase (GenBank:MAL515703), *Neurospora crassa* endoglucanase (GenBank:XM_324477), *Humicola insolens* endoglucanase V, *Myceliophthora thermophila* CBS 117.65 endoglucanase, *Thermoascus aurantiacus* endoglucanase I (GenBank:AF487830), *Trichoderma reesei* strain No. VTT-D-80133 endoglucanase (GenBank:M15665), and *Penicillium pinophilum* endoglucanase (WO 2012/062220).

Examples of cellobiohydrolases useful in the present invention include, but are not limited to, *Aspergillus aculeatus* cellobiohydrolase II (WO 2011/059740), *Aspergillus fumigatus* cellobiohydrolase I (WO 2013/028928), *Aspergillus fumigatus* cellobiohydrolase II (WO 2013/028928), *Chaetomium thermophilum* cellobiohydrolase I, *Chaetomium thermophilum* cellobiohydrolase II, *Humicola insolens* cellobiohydrolase I, *Myceliophthora thermophila* cellobiohydrolase II (WO 2009/042871), *Penicillium occitanis* cellobiohydrolase I (GenBank:AY690482), *Talaromyces emersonii* cellobiohydrolase I (GenBank:AF439936), *Thielavia hyrcanie* cellobiohydrolase II (WO 2010/141325), *Thielavia terrestris* cellobiohydrolase II (CEL6A, WO 2006/074435), *Trichoderma reesei* cellobiohydrolase I, *Trichoderma reesei* cellobiohydrolase II, and *Trichophaea saccata* cellobiohydrolase II (WO 2010/057086).

Examples of beta-glucosidases useful in the present invention include, but are not limited to, beta-glucosidases from *Aspergillus aculeatus* (Kawaguchi et al., 1996, Gene 173: 287-288), *Aspergillus fumigatus* (WO 2005/047499), *Aspergillus niger* (Dan et al., 2000, J. Biol. Chem. 275: 4973-4980), *Aspergillus oryzae* (WO 02/095014), *Penicillium brasilianum* IBT 20888 (WO 2007/019442 and WO 2010/088387), *Thielavia terrestris* (WO 2011/035029), and *Trichophaea saccata* (WO 2007/019442).

Other useful endoglucanases, cellobiohydrolases, and beta-glucosidases are disclosed in numerous Glycosyl Hydrolase families using the classification according to Henrissat, 1991, Biochem. J. 280: 309-316, and Henrissat and Bairoch, 1996, Biochem. J. 316: 695-696.

In the processes of the present invention, any AA9 polypeptide can be used as a component of the enzyme composition.

Examples of AA9 polypeptides useful in the processes of the present invention include, but are not limited to, AA9 polypeptides from *Thielavia terrestris* (WO 2005/074647, WO 2008/148131, and WO 2011/035027), *Thermoascus*

aurantiacus (WO 2005/074656 and WO 2010/065830), *Trichoderma reesei* (WO 2007/089290 and WO 2012/149344), *Myceliophthora thermophila* (WO 2009/085935, WO 2009/085859, WO 2009/085864, WO 2009/085868, and WO 2009/033071), *Aspergillus fumigatus* (WO 2010/138754), *Penicillium pinophilum* (WO 2011/005867), *Thermoascus* sp. (WO 2011/039319), *Penicillium* sp. (emersonii) (WO 2011/041397 and WO 2012/000892), *Thermoascus crustaceus* (WO 2011/041504), *Aspergillus aculeatus* (WO 2012/030799), *Thermomyces lanuginosus* (WO 2012/113340, WO 2012/129699, WO 2012/130964, and WO 2012/129699), *Aurantiporus alborubescens* (WO 2012/122477), *Trichophaea saccata* (WO 2012/122477), *Penicillium thomii* (WO 2012/122477), *Talaromyces stipitatus* (WO 2012/135659), *Humicola insolens* (WO 2012/146171), *Malbranchea cinnamomea* (WO 2012/101206), *Talaromyces leycettanus* (WO 2012/101206), *Chaetomium thermophilum* (WO 2012/101206), *Talaromyces thermophilus* (WO 2012/129697 and WO 2012/130950), *Acrophialophora fusispora* (WO 2013/043910), and *Corynascus sepedonium* (WO 2013/043910).

In one aspect, the AA9 polypeptide is used in the presence of a soluble activating divalent metal cation according to WO 2008/151043 or WO 2012/122518, e.g., manganese or copper.

In another aspect, the AA9 polypeptide is used in the presence of a dioxy compound, a bicyclic compound, a heterocyclic compound, a nitrogen-containing compound, a quinone compound, a sulfur-containing compound, or a liquor obtained from a pretreated cellulosic or hemicellulosic material such as pretreated corn stover (WO 2012/021394, WO 2012/021395, WO 2012/021396, WO 2012/021399, WO 2012/021400, WO 2012/021401, WO 2012/021408, and WO 2012/021410).

In one aspect, such a compound is added at a molar ratio of the compound to glucosyl units of cellulose of about 10^{-6} to about 10, e.g., about 10^{-6} to about 7.5, about 10^{-6} to about 5, about 10^{-6} to about 2.5, about 10^{-6} to about 1, about 10^{-5} to about 1, about 10^{-5} to about 10^{-1} , about 10^{-4} to about 10^{-1} , about 10^{-3} to about 10^{-1} , or about 10^{-3} to about 10^{-2} . In another aspect, an effective amount of such a compound is about 0.1 μ M to about 1 M, e.g., about 0.5 μ M to about 0.75 M, about 0.75 μ M to about 0.5 M, about 1 μ M to about 0.25 M, about 1 μ M to about 0.1 M, about 5 μ M to about 50 mM, about 10 μ M to about 25 mM, about 50 μ M to about 25 mM, about 10 μ M to about 10 mM, about 5 μ M to about 5 mM, or about 0.1 mM to about 1 mM.

The term "liquor" means the solution phase, either aqueous, organic, or a combination thereof, arising from treatment of a lignocellulose and/or hemicellulose material in a slurry, or monosaccharides thereof, e.g., xylose, arabinose, mannose, etc., under conditions as described in WO 2012/021401, and the soluble contents thereof. A liquor for cellulolytic enhancement of an AA9 polypeptide can be produced by treating a lignocellulose or hemicellulose material (or feedstock) by applying heat and/or pressure, optionally in the presence of a catalyst, e.g., acid, optionally in the presence of an organic solvent, and optionally in combination with physical disruption of the material, and then separating the solution from the residual solids. Such conditions determine the degree of cellulolytic enhancement obtainable through the combination of liquor and an AA9 polypeptide during hydrolysis of a cellulosic substrate by a cellulolytic enzyme preparation. The liquor can be separated from the treated material using a method standard in the art, such as filtration, sedimentation, or centrifugation.

In one aspect, an effective amount of the liquor to cellulose is about 10^{-6} to about 10 g per g of cellulose, e.g., about 10^{-6} to about 7.5 g, about 10^{-6} to about 5 g, about 10^{-6} to about 2.5 g, about 10^{-6} to about 1 g, about 10^{-5} to about 1 g, about 10^{-5} to about 10^{-1} g, about 10^{-4} to about 10^{-1} g, about 10^{-3} to about 10^{-1} g, or about 10^{-3} to about 10^{-2} g per g of cellulose.

In one aspect, the one or more hemicellulolytic enzymes comprise a commercial hemicellulolytic enzyme preparation. Examples of commercial hemicellulolytic enzyme preparations suitable for use in the present invention include, for example, SHEARZYME™ (Novozymes A/S), CELLIC® HTec (Novozymes A/S), CELLIC® HTec2 (Novozymes A/S), CELLIC® HTec3 (Novozymes A/S), VISCOZYME® (Novozymes A/S), ULTRAFLO® (Novozymes A/S), PULPZYME® HC (Novozymes A/S), MULTIFECT® Xylanase (Genencor), ACCELLERASE® XY (Genencor), ACCELLERASE® XC (Genencor), ECOPULP® TX-200A (AB Enzymes), HSP 6000 Xylanase (DSM), DEPOL™ 333P (Biocatalysts Limit, Wales, UK), DEPOL™ 740L (Biocatalysts Limit, Wales, UK), and DEPOL™ 762P (Biocatalysts Limit, Wales, UK), ALTERNA FUEL 100P (Dyadic), and ALTERNA FUEL 200P (Dyadic).

Examples of xylanases useful in the processes of the present invention include, but are not limited to, xylanases from *Aspergillus aculeatus* (GeneSeqP:AAR63790; WO 94/21785), *Aspergillus fumigatus* (WO 2006/078256), *Penicillium pinophilum* (WO 2011/041405), *Penicillium* sp. (WO 2010/126772), *Thermomyces lanuginosus* (GeneSeqP:BAA22485), *Talaromyces thermophilus* (GeneSeqP:BAA22834), *Thielavia terrestris* NRRL 8126 (WO 2009/079210), and *Trichophaea saccata* (WO 2011/057083).

Examples of acetylxylan esterases useful in the processes of the present invention include, but are not limited to, beta-xyllosidases from *Neurospora crassa* (SwissProt:Q7SOW4), *Trichoderma reesei* (UniProtKB/TrEMBL:Q92458), *Talaromyces emersonii* (SwissProt:Q8X212), and *Talaromyces thermophilus* (GeneSeqP:BAA22816).

Examples of acetylxylan esterases useful in the processes of the present invention include, but are not limited to, acetylxylan esterases from *Aspergillus aculeatus* (WO 2010/108918), *Chaetomium globosum* (UniProt:Q2GWX4), *Chaetomium gracile* (GeneSeqP:AAB82124), *Humicola insolens* DSM 1800 (WO 2009/073709), *Hypocrea jecorina* (WO 2005/001036), *Myceliophthora thermophila* (WO 2010/014880), *Neurospora crassa* (UniProt:q7s259), *Phaeosphaeria nodorum* (UniProt:QOUHJ1), and *Thielavia terrestris* NRRL 8126 (WO 2009/042846).

Examples of feruloyl esterases (ferulic acid esterases) useful in the processes of the present invention include, but are not limited to, feruloyl esterases from *Humicola insolens* DSM 1800 (WO 2009/076122), *Neosartorya fischeri* (UniProt:A1D9T4), *Neurospora crassa* (UniProt:Q9HGR3), *Penicillium aurantiogriseum* (WO 2009/127729), and *Thielavia terrestris* (WO 2010/053838 and WO 2010/065448).

Examples of arabinofuranosidases useful in the processes of the present invention include, but are not limited to, arabinofuranosidases from *Aspergillus niger* (GeneSeqP:AAR94170), *Humicola insolens* DSM 1800 (WO 2006/114094 and WO 2009/073383), and *M. giganteus* (WO 2006/114094).

Examples of alpha-glucuronidases useful in the processes of the present invention include, but are not limited to, alpha-glucuronidases from *Aspergillus clavatus* (UniProt:alcc12), *Aspergillus fumigatus* (SwissProt:Q4WW45),

Aspergillus niger (UniProt:Q96VVX9), *Aspergillus terreus* (SwissProt:Q0CJP9), *Humicola insolens* (WO 2010/014706), *Penicillium aurantiogriseum* (WO 2009/068565), *Talaromyces emersonii* (UniProt:Q8X211), and *Trichoderma reesei* (UniProt:Q99024).

Examples of oxidoreductases useful in the processes of the present invention include, but are not limited to, *Aspergillus lentilis* catalase, *Aspergillus fumigatus* catalase, *Aspergillus niger* catalase, *Aspergillus oryzae* catalase, *Humicola insolens* catalase, *Neurospora crassa* catalase, *Penicillium emersonii* catalase, *Scytalidium thermophilum* catalase, *Talaromyces stipitatus* catalase, *Thermoascus aurantiacus* catalase, *Coprinus cinereus* laccase, *Myceliophthora thermophila* laccase, *Polyporus pinsitus* laccase, *Pycnoporus cinnabarinus* laccase, *Rhizoctonia solani* laccase, *Streptomyces coelicolor* laccase, *Coprinus cinereus* peroxidase, Soy peroxidase, Royal palm peroxidase.

The polypeptides having enzyme activity used in the processes of the present invention may be produced by fermentation of the above-noted microbial strains on a nutrient medium containing suitable carbon and nitrogen sources and inorganic salts, using procedures known in the art (see, e.g., Bennett, J. W. and LaSure, L. (eds.), *More Gene Manipulations in Fungi*, Academic Press, C A, 1991). Suitable media are available from commercial suppliers or may be prepared according to published compositions (e.g., in catalogues of the American Type Culture Collection). Temperature ranges and other conditions suitable for growth and enzyme production are known in the art (see, e.g., Bailey, J. E., and Ollis, D.F., *Biochemical Engineering Fundamentals*, McGraw-Hill Book Company, NY, 1986).

The fermentation can be any method of cultivation of a cell resulting in the expression or isolation of an enzyme or protein. Fermentation may, therefore, be understood as comprising shake flask cultivation, or small- or large-scale fermentation (including continuous, batch, fed-batch, or solid-state fermentations) in laboratory or industrial fermentors performed in a suitable medium and under conditions allowing the enzyme to be expressed or isolated. The resulting enzymes produced by the methods described above may be recovered from the fermentation medium and purified by conventional procedures.

Fermentation. The fermentable sugars obtained from the hydrolyzed cellulosic or hemicellulosic material can be fermented by one or more fermenting microorganisms capable of fermenting the sugars directly or indirectly into a desired fermentation product.

"Fermentation" or "fermentation process" refers to any fermentation process or any process comprising a fermentation step. Fermentation processes also include fermentation processes used in the consumable alcohol industry (e.g., beer and wine), dairy industry (e.g., fermented dairy products), leather industry, and tobacco industry. The fermentation conditions depend on the desired fermentation product and fermenting organism and can easily be determined by one skilled in the art.

In the fermentation step, sugars, released from the cellulosic or hemicellulosic material as a result of the pretreatment and enzymatic hydrolysis steps, are fermented to a product, e.g., ethanol, by a fermenting organism, such as yeast. Hydrolysis (saccharification) and fermentation can be separate or simultaneous.

Any suitable hydrolyzed cellulosic or hemicellulosic material can be used in the fermentation step in practicing the present invention. The material is generally selected based on economics, i.e., costs per equivalent sugar potential, and recalcitrance to enzymatic conversion.

The term "fermentation medium" is understood herein to refer to a medium before the fermenting microorganism(s) is(are) added, such as, a medium resulting from a saccharification process, as well as a medium used in a simultaneous saccharification and fermentation process (SSF).

"Fermenting microorganism" refers to any microorganism, including bacterial and fungal organisms, suitable for use in a desired fermentation process to produce a fermentation product. The fermenting organism can be hexose and/or pentose fermenting organisms, or a combination thereof. Both hexose and pentose fermenting organisms are well known in the art. Suitable fermenting microorganisms are able to ferment, i.e., convert, sugars, such as glucose, xylose, xylulose, arabinose, maltose, mannose, galactose, and/or oligosaccharides, directly or indirectly into the desired fermentation product. Examples of bacterial and fungal fermenting organisms producing ethanol are described by Lin et al., 2006, *Appl. Microbiol. Biotechnol.* 69: 627-642.

Examples of fermenting microorganisms that can ferment hexose sugars include bacterial and fungal organisms, such as yeast. Yeast include strains of *Candida*, *Kluyveromyces*, and *Saccharomyces*, e.g., *Candida sonorensis*, *Kluyveromyces marxianus*, and *Saccharomyces cerevisiae*.

Examples of fermenting organisms that can ferment pentose sugars in their native state include bacterial and fungal organisms, such as some yeast. Xylose fermenting yeast include strains of *Candida*, preferably *C. sheatae* or *C. sonorensis*; and strains of *Pichia*, e.g., *P. stipitis*, such as *P. stipitis* CBS 5773. Pentose fermenting yeast include strains of *Pachysolen*, preferably *P. tannophilus*. Organisms not capable of fermenting pentose sugars, such as xylose and arabinose, may be genetically modified to do so by methods known in the art.

Examples of bacteria that can efficiently ferment hexose and pentose to ethanol include, for example, *Bacillus coagulans*, *Clostridium acetobutylicum*, *Clostridium thermocellum*, *Clostridium phytofermentans*, *Geobacillus* sp., *Thermoanaerobacter saccharolyticum*, and *Zymomonas mobilis* (Philippidis, G. P., 1996, Cellulose bioconversion technology, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, D.C., 179-212).

Other fermenting organisms include strains of *Bacillus*, such as *Bacillus coagulans*; *Candida*, such as *C. sonorensis*, *C. methanosorbosa*, *C. diddensiae*, *C. parapsilosis*, *C. nae-dodendra*, *C. blankii*, *C. entomophila*, *C. brassicae*, *C. pseudotropicalis*, *C. boidinii*, *C. utilis*, and *C. sheatae*; *Clostridium*, such as *C. acetobutylicum*, *C. thermocellum*, and *C. phytofermentans*; *E. coli*, especially *E. coli* strains that have been genetically modified to improve the yield of ethanol; *Geobacillus* sp.; *Hansenula*, such as *Hansenula anomala*; *Klebsiella*, such as *K. oxytoca*; *Kluyveromyces*, such as *K. marxianus*, *K. lactis*, *K. thermotolerans*, and *K. fragilis*; *Schizosaccharomyces*, such as *S. pombe*; *Thermoanaerobacter*, such as *Thermoanaerobacter saccharolyticum*; and *Zymomonas*, such as *Zymomonas mobilis*.

Commercially available yeast suitable for ethanol production include, e.g., BIO-FERM® AFT and XR (Lallemand Specialties, Inc., USA), ETHANOL RED® yeast (Lesaffre et Compagnie, France), FALI® (AB Mauri Food Inc., USA), FERMIOLE® (Rymco International AG, Denmark), GERT STRAND™ (Gert Strand AB, Sweden), and SUPER-START™ and THERMOSACC® fresh yeast (Lallemand Specialties, Inc., USA).

In an aspect, the fermenting microorganism has been genetically modified to provide the ability to ferment pen-

tose sugars, such as xylose utilizing, arabinose utilizing, and xylose and arabinose co-utilizing microorganisms.

The cloning of heterologous genes into various fermenting microorganisms has led to the construction of organisms capable of converting hexoses and pentoses to ethanol (co-fermentation) (Chen and Ho, 1993, *Appl. Biochem. Biotechnol.* 39-40: 135-147; Ho et al., 1998, *Appl. Environ. Microbiol.* 64: 1852-1859; Kotter and Ciriacy, 1993, *Appl. Microbiol. Biotechnol.* 38: 776-783; Walfridsson et al., 1995, *Appl. Environ. Microbiol.* 61: 4184-4190; Kuyper et al., 2004, *FEMS Yeast Research* 4: 655-664; Beall et al., 1991, *Biotech. Bioeng.* 38: 296-303; Ingram et al., 1998, *Biotechnol. Bioeng.* 58: 204-214; Zhang et al., 1995, *Science* 267: 240-243; Deanda et al., 1996, *Appl. Environ. Microbiol.* 62: 4465-4470; WO 03/062430).

In one aspect, the fermenting organism comprises a polynucleotide encoding a polypeptide having xylanase activity of the present invention.

In another aspect, the fermenting organism comprises one or more polynucleotides encoding one or more cellulolytic enzymes, hemicellulolytic enzymes, and accessory enzymes described herein.

It is well known in the art that the organisms described above can also be used to produce other substances, as described herein.

The fermenting microorganism is typically added to the degraded cellulosic or hemicellulosic material or hydrolysate and the fermentation is performed for about 8 to about 96 hours, e.g., about 24 to about 60 hours. The temperature is typically between about 26° C. to about 60° C., e.g., about 32° C. or 50° C., and about pH 3 to about pH 8, e.g., pH 4-5, 6, or 7.

In one aspect, the yeast and/or another microorganism are applied to the degraded cellulosic or hemicellulosic material and the fermentation is performed for about 12 to about 96 hours, such as typically 24-60 hours. In another aspect, the temperature is preferably between about 20° C. to about 60° C., e.g., about 25° C. to about 50° C., about 32° C. to about 50° C., or about 32° C. to about 50° C., and the pH is generally from about pH 3 to about pH 7, e.g., about pH 4 to about pH 7. However, some fermenting organisms, e.g., bacteria, have higher fermentation temperature optima. Yeast or another microorganism is preferably applied in amounts of approximately 10⁵ to 10¹², preferably from approximately 10⁷ to 10¹⁰, especially approximately 2x10⁸ viable cell count per ml of fermentation broth. Further guidance in respect of using yeast for fermentation can be found in, e.g., "The Alcohol Textbook" (Editors K. Jacques, T. P. Lyons and D. R. Kelsall, Nottingham University Press, United Kingdom 1999), which is hereby incorporated by reference.

A fermentation stimulator can be used in combination with any of the processes described herein to further improve the fermentation process, and in particular, the performance of the fermenting microorganism, such as, rate enhancement and ethanol yield. A "fermentation stimulator" refers to stimulators for growth of the fermenting microorganisms, in particular, yeast. Preferred fermentation stimulators for growth include vitamins and minerals. Examples of vitamins include multivitamins, biotin, pantothenate, nicotinic acid, meso-inositol, thiamine, pyridoxine, para-aminobenzoic acid, folic acid, riboflavin, and Vitamins A, B, C, D, and E. See, for example, Alfenore et al., Improving ethanol production and viability of *Saccharomyces cerevisiae* by a vitamin feeding strategy during fed-batch process, Springer-Verlag (2002), which is hereby incorporated by

reference. Examples of minerals include minerals and mineral salts that can supply nutrients comprising P, K, Mg, S, Ca, Fe, Zn, Mn, and Cu.

Fermentation products: A fermentation product can be any substance derived from the fermentation. The fermentation product can be, without limitation, an alcohol (e.g., arabinitol, n-butanol, isobutanol, ethanol, glycerol, methanol, ethylene glycol, 1,3-propanediol [propylene glycol], butanediol, glycerin, sorbitol, and xylitol); an alkane (e.g., pentane, hexane, heptane, octane, nonane, decane, undecane, and dodecane), a cycloalkane (e.g., cyclopentane, cyclohexane, cycloheptane, and cyclooctane), an alkene (e.g., pentene, hexene, heptene, and octene); an amino acid (e.g., aspartic acid, glutamic acid, glycine, lysine, serine, and threonine); a gas (e.g., methane, hydrogen (H₂), carbon dioxide (CO₂), and carbon monoxide (CO)); isoprene; a ketone (e.g., acetone); an organic acid (e.g., acetic acid, acetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, oxaloacetic acid, propionic acid, succinic acid, and xylonic acid); and polyketide.

In one aspect, the fermentation product is an alcohol. The term "alcohol" encompasses a substance that contains one or more hydroxyl moieties. The alcohol can be, but is not limited to, n-butanol, isobutanol, ethanol, methanol, arabinitol, butanediol, ethylene glycol, glycerin, glycerol, 1,3-propanediol, sorbitol, xylitol. See, for example, Gong et al., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Silveira and Jonas, 2002, *Appl. Microbiol. Biotechnol.* 59: 400-408; Nigam and Singh, 1995, *Process Biochemistry* 30(2): 117-124; Ezeji et al., 2003, *World Journal of Microbiology and Biotechnology* 19(6): 595-603.

In another aspect, the fermentation product is an alkane. The alkane can be an unbranched or a branched alkane. The alkane can be, but is not limited to, pentane, hexane, heptane, octane, nonane, decane, undecane, or dodecane.

In another aspect, the fermentation product is a cycloalkane. The cycloalkane can be, but is not limited to, cyclopentane, cyclohexane, cycloheptane, or cyclooctane.

In another aspect, the fermentation product is an alkene. The alkene can be an unbranched or a branched alkene. The alkene can be, but is not limited to, pentene, hexene, heptene, or octene.

In another aspect, the fermentation product is an amino acid. The organic acid can be, but is not limited to, aspartic acid, glutamic acid, glycine, lysine, serine, or threonine. See, for example, Richard and Margaritis, 2004, *Biotechnology and Bioengineering* 87(4): 501-515.

In another aspect, the fermentation product is a gas. The gas can be, but is not limited to, methane, H₂, CO₂, or CO. See, for example, Kataoka et al., 1997, *Water Science and Technology* 36(6-7): 41-47; and Gunaseelan, 1997, *Biomass and Bioenergy* 13(1-2): 83-114.

In another aspect, the fermentation product is isoprene.

In another aspect, the fermentation product is a ketone. The term "ketone" encompasses a substance that contains one or more ketone moieties. The ketone can be, but is not limited to, acetone.

In another aspect, the fermentation product is an organic acid. The organic acid can be, but is not limited to, acetic acid, acetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glu-

caric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, propionic acid, succinic acid, or xylonic acid. See, for example, Chen and Lee, 1997, *Appl. Biochem. Biotechnol.* 63-65: 435-448.

In another aspect, the fermentation product is polyketide.

Recovery. The fermentation product(s) can be optionally recovered from the fermentation medium using any method known in the art including, but not limited to, chromatography, electrophoretic procedures, differential solubility, distillation, or extraction. For example, alcohol is separated from the fermented cellulosic or hemicellulosic material and purified by conventional methods of distillation. Ethanol with a purity of up to about 96 vol. can be obtained, which can be used as, for example, fuel ethanol, drinking ethanol, i.e., potable neutral spirits, or industrial ethanol.

Signal Peptide

The present invention also relates to an isolated polynucleotide encoding a signal peptide comprising or consisting of amino acids 1 to 19 of SEQ ID NO: 2. The polynucleotide may further comprise a gene encoding a protein, which is operably linked to the signal peptide. The protein is preferably foreign to the signal peptide. In one aspect, the polynucleotide encoding the signal peptide is nucleotides 1 to 107 of SEQ ID NO: 1.

The present invention also relates to nucleic acid constructs, expression vectors and recombinant host cells comprising such a polynucleotide.

The present invention also relates to methods of producing a protein, comprising (a) cultivating a recombinant host cell comprising such a polynucleotide; and optionally (b) recovering the protein.

The protein may be native or heterologous to a host cell. The term "protein" is not meant herein to refer to a specific length of the encoded product and, therefore, encompasses peptides, oligopeptides, and polypeptides. The term "protein" also encompasses two or more polypeptides combined to form the encoded product. The proteins also include hybrid polypeptides and fused polypeptides.

Preferably, the protein is a hormone, enzyme, receptor or portion thereof, antibody or portion thereof, or reporter. For example, the protein may be a hydrolase, isomerase, ligase, lyase, oxidoreductase, or transferase, e.g., an alpha-galactosidase, alpha-glucosidase, aminopeptidase, amylase, betagalactosidase, beta-glucosidase, beta-xylosidase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, glucoamylase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, or xylanase.

The gene may be obtained from any prokaryotic, eukaryotic, or other source.

The present invention is further described by the following examples that should not be construed as limiting the scope of the invention.

EXAMPLES

Example 1: Cloning and Expression of a Genomic *Neosartorya Massa* GH10 Xylanase

The xylanase gene was obtained from *Neosartorya massa* strain CBS 117265, which was isolated from a soil sample collected in Denmark.

Genomic DNA was isolated from *Neosartorya massa* strain CBS 117265 (CBS-KNAW Fungal Biodiversity Centre) using a FASTDNA® Spin Kit for Soil (MP Biomedicals) and submitted to whole genome sequencing by standard methods. The whole genome sequence was assembled with a SPAdes v3.1.1 genome assembler (Bankevich et al., 2012, *J. Comput. Biol.* 19(5): 455-477), and genes were annotated with the GeneMark 2.3c gene prediction software (Ter-Hovhannisyan et al., 2008, *Genome Res.* 18(12): 1979-1990).

The *Neosartorya massa* strain CBS 117265 genome assembly sequences were analyzed for xylanases using the CAZY database GH10 family (Lombard et al., 2014, The Carbohydrate-active enzymes database (CAZy) in 2013. *Nucleic Acids Research* 42: D490-D495). This analysis identified a gene encoding a GH10 xylanase which was subsequently cloned and recombinantly expressed in *Aspergillus oryzae*.

The gene encoding the GH10 xylanase was cloned by PCR from the genomic DNA using gene-specific primers shown below that also append a Kozak translation initiation sequence "TCCACC" immediately 5' of the start codon.

Primer F-NmasGH10:

(SEQ ID NO: 4)

5' - ACACACTGGGATCCACCATG**GGTTCATCTTTCTTCAATCGCAGC**-

3'

Primer R-NmasGH10:

(SEQ ID NO: 5)

5' - CCCTCTAGATCTCGAGTTACAGACACTGCGAGTACCAATC-3'

Bold letters represent gene sequences. The underlined sequences are homologous to the insertion sites of pDau109 (WO 2005/042735).

An MJ Research PTC-200 DNA engine (MJ Research Inc.) was used to perform the PCR. A PHUSION® High-Fidelity PCR Kit (Finnzymes Oy) was used for the amplification. The PCR was composed of 5 µl of 5× HF buffer (Finnzymes Oy), 0.5 µl of dNTPs (10 mM), 0.5 µl of PHUSION® DNA polymerase (0.2 units/µl) (Finnzymes Oy), 5 µM of each primer, 0.5 µl of *Neosartorya massa* genomic DNA (100 ng/µl), and 16.5 µl of deionized water in a total volume of 25 µl. The PCR conditions were 1 cycle at 98° C. for 2 minutes; 35 cycles each at 98° C. for 10 seconds and 72° C. for 2.5 minutes; and 1 cycle at 72° C. for 10 minutes. The sample was then held at 12° C. until removed from the PCR machine.

The reaction product was isolated by 1.0% agarose gel electrophoresis using TAE buffer composed of 4.84 g of Tris base, 1.14 ml of glacial acetic acid, 2 ml of 0.5 M EDTA pH 8.0, and deionized water to 1 liter where a product band of approximately 1480 bp was excised from the gel and purified using an ILLUSTRA™ GFX™ PCR DNA and Gel Band Purification Kit (GE Healthcare Life Sciences) according to the manufacturers instructions. The fragment was then cloned into Bam HI and Xho I digested pDau109 using an IN-FUSION™ Cloning Kit resulting in the plasmid pNmasGH10. Cloning of the gene into Bam HI-Xho I digested pDau109 resulted in transcription of the *Neosartorya massa* GH10 gene under the control of a NA2-tpi double promoter. NA2-tpi is a modified promoter from the gene encoding the *Aspergillus niger* neutral alpha-amylase in which the untranslated leader has been replaced by an untranslated leader from the gene encoding the *Aspergillus nidulans* triose phosphate isomerase.

The cloning protocol was performed according to the IN-FUSION™ Cloning Kit instructions generating the GH10 xylanase construct. The treated plasmid and insert was transformed into ONE SHOT® TOP10F® Chemically Competent *E. coli* cells (Invitrogen) according to the manufacturer's protocol and spread onto LB plates supplemented with 0.1 mg of ampicillin per ml. LB plates were composed of 10 g of tryptone, 5 g of yeast extract, 5 g of sodium chloride, 15 g of Bacto agar, and deionized water to 1 liter. After incubation at 37° C. overnight, colonies were observed growing under selection on the LB ampicillin plates. Colonies of the transformation were cultivated in LB medium supplemented with 0.1 mg of ampicillin per ml and plasmids were isolated with a QIAPREP® Spin Miniprep Kit (QIAGEN Inc.) according to the manufacturers protocol. LB medium was composed of 10 g of tryptone, 5 g of yeast extract, 5 g of sodium chloride, and deionized water to 1 liter.

Isolated plasmids were sequenced with vector primers and gene specific primers in order to determine a representative plasmid expression clone that was free of PCR errors, and a plasmid without errors was selected for expression of the polypeptide. The expression construct was transformed into *Aspergillus oryzae* strain MT3568 (WO 2011/057140) to produce the secreted mature polypeptide of SEQ ID NO: 2. Transformants were selected on acetamide during regeneration from protoplasts and subsequently re-isolated under selection (Christensen et al., 1988, *Biotechnology* 6: 1419-1422 and WO 2004/032648).

For production of the recombinant xylanase, a single *Aspergillus oryzae* transformant was cultured in twenty 500 ml baffled flasks each containing 150 ml of DAP-4C-1 medium (WO 2012/103350). The cultures were shaken on a rotary table at 150 rpm for 4 days at 30° C. The culture broth was subsequently separated from cellular material by passage through a 0.22 µm filter.

Example 2: Characterization of the Genomic DNA from *Neosartorya Massa* Encoding a GH10 Xylanase

The genomic DNA sequence and deduced amino acid sequence of the GH10 polypeptide coding sequence from *Neosartorya massa* are shown in SEQ ID NO: 1 and SEQ ID NO: 2, respectively. The genomic DNA coding sequence is 1418 bp including the stop codon interrupted by 4 introns of 50 bp (nucleotides 48 to 97), 63 bp (nucleotides 255 to 317), 47 bp (nucleotides 451 to 497), and 58 bp (nucleotides 612 to 669). Using the SignalP 4.0 program (Petersen et al., 2011, *Nature Methods* 8: 785-786), a signal peptide of 19 amino acids was predicted. The predicted mature polypeptide contains 380 amino acids with a predicted molecular mass of 41 kDa and a predicted isoelectric point of 5.6. The GH10 catalytic domain extends from approximately amino acids 20 to 341 of the full-length polypeptide. A family 1 carbohydrate binding module extends from approximately amino acids 363 to 399.

A comparative pairwise global alignment of amino acid sequences was determined using the Needleman and Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) with a gap open penalty of 10, a gap extension penalty of 0.5, and the EBLOSUM62 matrix. The alignment showed that the deduced amino acid sequence of the *Neosartorya massa* genomic DNA encoding a GH10 polypeptide shares 98% identity (excluding gaps) to the deduced

amino acid sequence of a GH10 polypeptide from *Aspergillus turcosus* (SWISSPROT:A0A229X5W7).

Example 3: Purification of the Recombinant
Neosartorya Massa GH10 Xylanase

Filtered broth (Example 1) was adjusted to pH 7.0 and filtered using a 0.22 μ m PES filter (Nalge Nunc International). Ammonium sulfate was added to the filtrate to a concentration of 1.8 M and then loaded onto a PHENYL SEPHAROSE™ 6 Fast Flow column (high sub) (GE Healthcare) equilibrated with 1.8 M ammonium sulfate, 25 mM HEPES pH 7.0. The bound protein was eluted with 1.0 M ammonium sulfate, 25 mM HEPES pH 7.0. Fractions were collected, analyzed by SDS-PAGE using a NUPAGE® 4-12% Bis-Tris SDS-PAGE gel with 2-(N-morpholino)ethanesulfonic acid (MES) running buffer according to the manufacturers instructions (Thermo Fischer Scientific), and pooled based on the presence of the recombinant xylanase protein. The pooled fractions were applied to a SEPHADAX™ G-25 (medium) (GE Healthcare) column equilibrated in 25 mM HEPES pH 7.0. Fractions were collected, analyzed by SDS-PAGE as above, and pooled based on the presence of the recombinant xylanase protein. The pooled fractions were applied to a SOURCE™ 15Q (GE Healthcare) column equilibrated in 25 mM HEPES pH 7.0 and the bound proteins were eluted with a linear gradient from 0-1000 mM sodium chloride over 20 column volumes (CV). Fractions were collected, analyzed by SDS-PAGE as above, and pooled based on the presence of the recombinant xylanase protein.

Example 4: Pretreated Wheat Straw (PWS)
Hydrolysis Assay

Wheat straw was pretreated by steam explosion and the unwashed slurry was adjusted to pH 5.0 using 1.0 N sodium hydroxide.

A 96-well plate was constructed by machining a TEF-LON® plate of depth ¼ inch with 96 cone-shaped wells, diameter ¼ inch at the upper surface and diameter ⅛ inch at the lower surface. The center of each well was at an equivalent position to the center of a corresponding well in a standard 96-well microtiter plate, approximately 23/64 inch on center. The resulting volume of each well was approximately 135 μ l. This 96-well TEF-LON® plate is hereinafter referred to as the "fill plate". The pH-adjusted pretreated wheat straw (PWS) was added to the holes in the fill plate by applying a suitable volume of the wheat straw to the upper surface of the plate, then using a spatula to spread the material over the surface and into the holes. Holes were deemed sufficiently full when wheat straw extruded through the hole in the bottom surface, forming noodle-like tubes. A MULTISCREEN® Column Loader Scraper (Millipore) held perpendicular to the fill plate surface was used to scrape excess wheat straw from the top and bottom surfaces of the fill plate, leaving the surfaces of the wheat straw in each well flush with the surfaces of the fill plate. The fill plate was then placed on the top of a 2.2 ml deep-well plate (Axygen) with the top surface adjacent to the open end of the well plate (e.g., the top of the well plate), and the wells aligned with the wheat straw-filled holes in the fill plate. The fill plate was secured in this position, and the assembly centrifuged at 2500 rpm (1350 $\times$ g) for 5 minutes in a SORVALL™ LEG-END™ RT+ (Thermo Scientific) to transfer the wheat straw to the deep-well plate. A 3 mm glass bead (Fisher Scientific) was placed in each well for mixing.

Hydrolysis of the PWS was conducted in a total reaction volume of 200-250 μ l. The hydrolysis was performed with 150 mg of insoluble PWS solids containing 50 mM sodium acetate pH 5.0 buffer and protein loadings of various enzyme compositions (expressed as mg protein per gram of cellulose). The enzyme compositions were added simultaneously to all wells in a volume ranging from 20 μ l to 50 μ l, for a final volume of 200-250 μ l in each reaction. The plate was then sealed using an ALPS-300™ plate heat sealer (Abgene), mixed thoroughly, and incubated at a specific temperature for 72 hours.

Following hydrolysis, samples were diluted in the assay plate with Milli-Q water and filtered using a 0.45 μ m MULTISCREEN® 96-well filtration plate (Millipore). Filtrates were analyzed by HPLC for glucose content as described below. When not used immediately, filtered aliquots were frozen at -20°C. The glucose concentrations of the diluted samples were measured using a 4.6 $\times$ 250 mm AMINEX® HPX-87H column (Bio-Rad Laboratories, Inc.) by elution with 0.005 M H₂SO₄ with 0.05% w/w benzoic acid, at 65° C. at a flow rate of 0.6 ml per minute, and quantitation by integration of the glucose signal from refractive index detection (CHEMSTATION, AGILENT 1100 HPLC, Agilent Technologies) calibrated by pure glucose standards.

The measured glucose concentrations were adjusted for the appropriate dilution factor. The net concentrations of enzymatically produced glucose from PWS were determined by adjusting the measured glucose concentration for corresponding background glucose concentration in PWS at zero time-point. All HPLC data processing was performed using MICROSOFT EXCEL™ software (Microsoft).

Example 5: Preparation of an Enzyme Composition
without GH10 Xylanase

The *Aspergillus fumigatus* GH10 xylanase (xyn3) (GENESEQP:AZ104884) was prepared recombinantly according to WO 2006/078256 using *Aspergillus oryzae* BECh2 (WO 2000/39322) as a host. Ammonium sulfate was added to filtered broth at a final concentration of 1.0 M and adjusted to pH 7.5 using 50% w/w sodium hydroxide. The protein was loaded onto a PHENYL SEPHAROSE™ HP column (GE Healthcare) equilibrated in 20 mM Tris pH 7.5 with 1.0 M ammonium sulfate, and bound proteins were eluted with 20 mM Tris pH 7.5 with no ammonium sulfate. Fractions were analyzed by 8-16% Tris-HCl SDS-PAGE gels (Bio-Rad Laboratories, Inc.), and pooled. The pooled protein was concentrated and buffer exchanged into 50 mM sodium acetate with 150 mM sodium chloride pH 5.0 using a VIVACELL® 100 10 kDa molecular weight cut-off ultra-filtration unit (Sartorius).

The *Neosartorya massa* GH10 xylanase was prepared as described in Example 3.

The *Talaromyces emersonii* CBS 393.64 beta-xylosidase (GENESEQP:AZ104896) was prepared recombinantly according to Rasmussen et al., 2006, *Biotechnology and Bioengineering* 94: 869-876 using *Aspergillus oryzae* JaL355 as a host (WO 2003/070956). Filtered broth was diluted 10-fold into 20 mM Tris pH 8.0 and adjusted to pH 8.0 using 1 M NaOH. The protein was loaded onto a Q SEPHAROSE® High Performance column (GE Healthcare) equilibrated in 20 mM Tris pH 8.0 and bound proteins were eluted with 20 mM Tris pH 8.0 with 1 M sodium chloride. Fractions were analyzed by SDS-PAGE using an 8-16% Tris-HCl gel (Bio-Rad Laboratories, Inc.), and pooled. The pooled protein was concentrated and buffer exchanged into

50 mM sodium acetate with 150 mM sodium chloride pH 5.0 using a VIVACELL™ 100 10 kDa molecular weight cut-off ultrafiltration unit.

The protein concentration for each of the purified mono components described above was calculated using the absorbance at 280 nm and the corresponding proteins extinction coefficient as determined by sequence.

A base enzyme composition without GH10 xylanase was prepared by adding the purified *Talaromyces emersonii* beta-xylosidase to a *Trichoderma reesei* composition composed of *Talaromyces leycettanus* cellobiohydrolase I, *Talaromyces leycettanus* cellobiohydrolase II, *Talaromyces emersonii* beta-xylosidase, *Aspergillus fumigatus* beta-glucosidase variant, *Penicillium emersonii* AA9 polypeptide, *Trichoderma reesei* endoglucanase I, and *Trichoderma reesei* endoglucanase II. The protein concentration of the *Trichoderma reesei* composition was measured using a BCA Protein Assay Kit (Pierce). The *Trichoderma reesei* composition was composed of approximately 20% *Talaromyces leycettanus* cellobiohydrolase I, 18% *Talaromyces leycettanus* cellobiohydrolase II, 5% *Talaromyces emersonii* beta-xylosidase, 8% *Aspergillus fumigatus* beta-glucosidase variant, and 15% *Penicillium emersonii* AA9 polypeptide, 8% *Trichoderma reesei* endoglucanase I and 8% *Trichoderma reesei* endoglucanase II based on protein. The enzyme composition is designated herein as “cellulolytic enzyme composition”.

Example 6: Comparison of the Effect of
Neosartorya Massa GH10 Xylanase and
Aspergillus fumigatus GH10 Xylanase on the
Hydrolysis of Unwashed PWS

The effect of *Neosartorya massa* GH10 xylanase on the hydrolysis of unwashed PWS as a substrate by the cellulolytic enzyme composition (Example 5) was determined at 50° C., 55° C. and 60° C. and compared to the cellulolytic enzyme composition without GH10 xylanase or with *Aspergillus fumigatus* GH10 xylanase. GH10 xylanase was added individually at 0.50 mg enzyme protein per g cellulose to 10.0 mg enzyme protein of the cellulase enzyme composition without GH10 per g cellulose. The reactions with unwashed PWS (15% total solids) were conducted for 72 hours at the desired temperature in 50 mM sodium acetate pH 5.0 buffer. All reactions were performed in 8 replicate reactions with shaking at 200 rpm throughout the hydrolysis.

The results shown in FIG. 1 demonstrated the effect of adding the *Neosartorya massa* GH10 xylanase to the cellulolytic enzyme composition without GH10 compared to the *Aspergillus fumigatus* GH10 xylanase at 50° C., 55° C., and 60° C. The increase in glucose concentration from addition of the *Neosartorya massa* GH10 xylanase compared to the *Aspergillus fumigatus* GH10 xylanase was displayed as a significant difference outside of the standard deviations measured for replicate reactions.

The invention is further defined by the following numbered paragraphs:

Paragraph 1. An isolated polypeptide having xylanase activity, selected from the group consisting of: (a) a polypeptide having at least 99% sequence identity to the mature polypeptide of SEQ ID NO: 2; (b) a polypeptide encoded by a polynucleotide having at least 99% sequence identity to the mature polypeptide coding sequence of SEQ ID NO: 1; (c) a variant of the mature polypeptide of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and (d) a fragment of the polypeptide of (a), (b), or (c) that has xylanase activity.

Paragraph 2. The polypeptide of paragraph 1, having at least 99% or 100% sequence identity to the mature polypeptide of SEQ ID NO: 2.

Paragraph 3. The polypeptide of paragraph 1 or 2, which is encoded by a polynucleotide having at least 99% or 100% sequence identity to the mature polypeptide coding sequence of SEQ ID NO: 1.

Paragraph 4. The polypeptide of any one of paragraphs 1-3, comprising or consisting of SEQ ID NO: 2, the mature polypeptide of SEQ ID NO: 2, or SEQ ID NO: 3.

Paragraph 5. The polypeptide of paragraph 4, wherein the mature polypeptide is amino acids 20 to 399 of SEQ ID NO: 2.

Paragraph 6. The polypeptide of any one of paragraphs 1-3, which is a variant of the mature polypeptide of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions.

Paragraph 7. The polypeptide of paragraph 1, which is a fragment of SEQ ID NO: 2, wherein the fragment has xylanase activity.

Paragraph 8. An isolated polypeptide comprising a catalytic domain selected from the group consisting of: (a) a catalytic domain having at least 99% or 100% sequence identity to amino acids 20 to 341 of SEQ ID NO: 2; (b) a catalytic domain encoded by a polynucleotide having at least 99% or 100% sequence identity to nucleotides 108 to 1241 of SEQ ID NO: 1; (c) a variant of amino acids 20 to 341 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and (d) a fragment of the catalytic domain of (a), (b), or (c) that has xylanase activity.

Paragraph 9. The polypeptide of paragraph 8, further comprising a carbohydrate binding module.

Paragraph 10. An isolated polypeptide comprising a carbohydrate binding module operably linked to a catalytic domain, wherein the binding domain is selected from the group consisting of: (a) a carbohydrate binding module having at least 99% or 100% sequence identity to amino acids 363 to 399 of SEQ ID NO: 2; (b) a carbohydrate binding module encoded by a polynucleotide having at least 99% or 100% sequence identity to nucleotides 1305 to 1415 of SEQ ID NO: 1; (c) a variant of amino acids 363 to 399 of SEQ ID NO: 2 comprising a substitution, deletion, and/or insertion at one or more positions; and (d) a fragment of (a), (b), or (c) that has carbohydrate binding activity.

Paragraph 11. The polypeptide of paragraph 10, wherein the catalytic domain is obtained from a hydrolase, isomerase, ligase, lyase, oxidoreductase, or transferase, e.g., an aminopeptidase, amylase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, alpha-galactosidase, beta-galactosidase, glucoamylase, alpha-glucosidase, beta-glucosidase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, xylanase, or beta-xylosidase.

Paragraph 12. A composition comprising the polypeptide of any one of paragraphs 1-11.

Paragraph 13. A whole broth formulation or cell culture composition comprising the polypeptide of any one of paragraphs 1-11.

Paragraph 14. An isolated polynucleotide encoding the polypeptide of any one of paragraphs 1-11.

Paragraph 15. A nucleic acid construct or expression vector comprising the polynucleotide of paragraph 14 operably linked to one or more control sequences that direct the production of the polypeptide in an expression host.

Paragraph 16. A recombinant host cell transformed with the polynucleotide of paragraph 14 operably linked to one or more control sequences that direct the production of the polypeptide.

Paragraph 17. A method of producing the polypeptide of any one of paragraphs 1-11, comprising cultivating a wild-type cell under conditions conducive for production of the polypeptide.

Paragraph 18. The method of paragraph 17, further comprising recovering the polypeptide.

Paragraph 19. A method of producing a polypeptide having xylanase activity, comprising cultivating the recombinant host cell of paragraph 16 under conditions conducive for production of the polypeptide.

Paragraph 20. The method of paragraph 19, further comprising recovering the polypeptide.

Paragraph 21. A transgenic plant, plant part or plant cell transformed with a polynucleotide encoding the polypeptide of any one of paragraphs 1-11.

Paragraph 22. A method of producing a polypeptide having xylanase activity, comprising cultivating the transgenic plant or plant cell of paragraph 21 under conditions conducive for production of the polypeptide.

Paragraph 23. The method of paragraph 22, further comprising recovering the polypeptide.

Paragraph 24. An isolated polynucleotide encoding a signal peptide comprising or consisting of amino acids 1 to 19 of SEQ ID NO: 2.

Paragraph 25. A nucleic acid construct or expression vector comprising a gene encoding a protein operably linked to the polynucleotide of paragraph 24, wherein the gene is foreign to the polynucleotide encoding the signal peptide.

Paragraph 26. A recombinant host cell comprising a gene encoding a protein operably linked to the polynucleotide of paragraph 24, wherein the gene is foreign to the polynucleotide encoding the signal peptide.

Paragraph 27. A method of producing a protein, comprising cultivating a recombinant host cell comprising a gene encoding a protein operably linked to the polynucleotide of paragraph 24, wherein the gene is foreign to the polynucleotide encoding the signal peptide, under conditions conducive for production of the protein.

Paragraph 28. The method of paragraph 27, further comprising recovering the protein.

Paragraph 29. A process for degrading a cellulosic or hemicellulosic material, comprising: treating the cellulosic or hemicellulosic material with an enzyme composition comprising the polypeptide having xylanase activity of any one of paragraphs 1-11.

Paragraph 30. The process of paragraph 29, wherein the cellulosic or hemicellulosic material is pretreated.

Paragraph 31. The process of paragraph 29 or 30, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

Paragraph 32. The process of paragraph 31, wherein the cellulase is one or more enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

Paragraph 33. The process of paragraph 31, wherein the hemicellulase is one or more enzymes selected from the group consisting of a xylanase, an acetylxytan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase.

Paragraph 34. The process of any one of paragraphs 29-33, further comprising recovering the degraded cellulosic or hemicellulosic material.

Paragraph 35. The process of paragraph 34, wherein the degraded cellulosic or hemicellulosic material is a sugar.

Paragraph 36. The process of paragraph 35, wherein the sugar is selected from the group consisting of glucose, xylose, mannose, galactose, and arabinose.

Paragraph 37. A process for producing a fermentation product, comprising: (a) saccharifying a cellulosic or hemicellulosic material with an enzyme composition comprising the polypeptide having xylanase activity of any one of paragraphs 1-11; (b) fermenting the saccharified cellulosic or hemicellulosic material with one or more fermenting microorganisms to produce the fermentation product; and (c) recovering the fermentation product from the fermentation.

Paragraph 38. The process of paragraph 37, wherein the cellulosic or hemicellulosic material is pretreated.

Paragraph 39. The process of paragraph 37 or 38, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

Paragraph 40. The process of paragraph 39, wherein the cellulase is one or more enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

Paragraph 41. The process of paragraph 39, wherein the hemicellulase is one or more enzymes selected from the group consisting of a xylanase, an acetylxytan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase.

Paragraph 42. The process of any one of paragraphs 37-41, wherein steps (a) and (b) are performed simultaneously in a simultaneous saccharification and fermentation.

Paragraph 43. The process of any one of paragraphs 37-42, wherein the fermentation product is an alcohol, an alkane, a cycloalkane, an alkene, an amino acid, a gas, isoprene, a ketone, an organic acid, or polyketide.

Paragraph 44. A process of fermenting a cellulosic or hemicellulosic material, comprising: fermenting the cellulosic or hemicellulosic material with one or more fermenting microorganisms, wherein the cellulosic or hemicellulosic material is saccharified with an enzyme composition comprising the polypeptide having xylanase activity of any one of paragraphs 1-11.

Paragraph 45. The process of paragraph 44, wherein the cellulosic or hemicellulosic material is pretreated before saccharification.

Paragraph 46. The process of paragraph 44 or 45, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

Paragraph 47. The process of paragraph 46, wherein the cellulase is one or more enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

Paragraph 48. The process of paragraph 46, wherein the hemicellulase is one or more enzymes selected from the group consisting of a xylanase, an acetylxytan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase.

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Paragraph 49. The process of any one of paragraphs 44-48, wherein the fermenting of the cellulosic or hemicellulosic material produces a fermentation product.

Paragraph 50. The process of paragraph 49, further comprising recovering the fermentation product from the fermentation.

Paragraph 51. The process of paragraph 49 or 50, wherein the fermentation product is an alcohol, an alkane, a cycloalkane, an alkene, an amino acid, a gas, isoprene, a ketone, an organic acid, or polyketide.

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The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control.

SEQUENCE LISTING

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<211> LENGTH: 1418

<212> TYPE: DNA

<213> ORGANISM: *Neosartorya massa*

<400> SEQUENCE: 1

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ccactggctg tgtgggcgtc accatctggg actggaccga caagtactcc tgggtcccca    1140
gcgtgttctc aggtcacggc gccgcattgc cttgggtaga gaactatgtg aagaagccag    1200
catacaatgg cattttggcg gggcttgagg caagcggctc cacgatgacc actaccttga    1260
ctaccaccac gaagacctct actacgacaa ctgcgacgtc tactggagtc getcagaaat    1320
ggggacagtg cggcgggtatc ggctggaccg gcccaacaac ctgcgtcagt gggactacct    1380
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<210> SEQ ID NO 2

<211> LENGTH: 399

<212> TYPE: PRT

<213> ORGANISM: *Neosartorya massa*

<400> SEQUENCE: 2

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Met Val His Leu Ser Ser Ile Ala Ala Ala Leu Thr Ala Leu Pro Leu
1          5          10          15

Val Tyr Gly Ala Gly Leu Asn Thr Ala Ala Lys Ala Lys Gly Leu Lys
          20          25          30

Tyr Phe Gly Ser Ala Thr Asp Asn Pro Glu Leu Thr Asp Thr Ala Tyr
          35          40          45

Val Ala Gln Leu Ser Asn Thr Asp Asp Phe Gly Gln Ile Thr Pro Gly
50          55          60

Asn Ser Met Lys Trp Asp Ala Thr Glu Pro Ser Gln Asn Ser Phe Ser
65          70          75          80

Phe Ala Asn Gly Asp Ala Ile Val Asn Leu Ala Asn Lys Asn Gly Gln
          85          90          95

Met Met Arg Cys His Thr Leu Val Trp His Ser Gln Leu Pro Asn Trp
          100          105          110

Val Ser Ser Gly Ser Trp Thr Asn Ala Thr Leu Leu Ala Ala Met Lys
          115          120          125

Asn His Ile Thr Asn Val Val Thr His Tyr Lys Gly Lys Cys Tyr Ala
130          135          140

Trp Asp Val Val Asn Glu Ala Leu Asn Glu Asp Gly Thr Phe Arg Asn
145          150          155          160

Ser Ile Phe Tyr Gln Thr Ile Gly Glu Ala Tyr Ile Pro Ile Ala Phe
          165          170          175

Ala Thr Ala Ala Ala Ala Asp Pro Ser Val Lys Leu Tyr Tyr Asn Asp
          180          185          190

Tyr Asn Ile Glu Tyr Ser Gly Ala Lys Ala Thr Ala Ala Gln Asn Ile
          195          200          205

Val Lys Met Ile Lys Ala Tyr Gly Ala Lys Ile Asp Gly Val Gly Leu
210          215          220

Gln Gly His Phe Ile Leu Gly Ser Thr Pro Ser Gln Ser Ala Leu Thr
225          230          235          240

Ser Val Leu Glu Gly Tyr Thr Ala Leu Gly Val Glu Val Ala Tyr Thr
          245          250          255

Glu Leu Asp Ile Arg Met Gln Leu Pro Ser Thr Asp Ala Lys Leu Ala
          260          265          270

Gln Gln Ala Thr Glu Tyr Gln Asn Val Ala Ala Ala Cys Val Gly Thr
          275          280          285

Thr Gly Cys Val Gly Val Thr Ile Trp Asp Trp Thr Asp Lys Tyr Ser
290          295          300

Trp Val Pro Ser Val Phe Ser Gly Tyr Gly Ala Ala Leu Pro Trp Asp
305          310          315          320

Glu Asn Tyr Val Lys Lys Pro Ala Tyr Asn Gly Ile Leu Ala Gly Leu
          325          330          335

Gly Ala Ser Gly Ser Thr Met Thr Thr Thr Leu Thr Thr Thr Thr Lys
          340          345          350

Thr Ser Thr Thr Thr Thr Ala Thr Ser Thr Gly Val Ala Gln Lys Trp
          355          360          365

Gly Gln Cys Gly Gly Ile Gly Trp Thr Gly Pro Thr Thr Cys Val Ser
          370          375          380

Gly Thr Thr Cys Gln Lys Leu Asn Asp Trp Tyr Ser Gln Cys Leu
385          390          395

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<210> SEQ ID NO 3

<211> LENGTH: 380

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<212> TYPE: PRT

<213> ORGANISM: Neosartorya massa

<400> SEQUENCE: 3

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 Ser Ala Thr Asp Asn Pro Glu Leu Thr Asp Thr Ala Tyr Val Ala Gln
 20 25 30
 Leu Ser Asn Thr Asp Asp Phe Gly Gln Ile Thr Pro Gly Asn Ser Met
 35 40 45
 Lys Trp Asp Ala Thr Glu Pro Ser Gln Asn Ser Phe Ser Phe Ala Asn
 50 55 60
 Gly Asp Ala Ile Val Asn Leu Ala Asn Lys Asn Gly Gln Met Met Arg
 65 70 75 80
 Cys His Thr Leu Val Trp His Ser Gln Leu Pro Asn Trp Val Ser Ser
 85 90 95
 Gly Ser Trp Thr Asn Ala Thr Leu Leu Ala Ala Met Lys Asn His Ile
 100 105 110
 Thr Asn Val Val Thr His Tyr Lys Gly Lys Cys Tyr Ala Trp Asp Val
 115 120 125
 Val Asn Glu Ala Leu Asn Glu Asp Gly Thr Phe Arg Asn Ser Ile Phe
 130 135 140
 Tyr Gln Thr Ile Gly Glu Ala Tyr Ile Pro Ile Ala Phe Ala Thr Ala
 145 150 155 160
 Ala Ala Ala Asp Pro Ser Val Lys Leu Tyr Tyr Asn Asp Tyr Asn Ile
 165 170 175
 Glu Tyr Ser Gly Ala Lys Ala Thr Ala Ala Gln Asn Ile Val Lys Met
 180 185 190
 Ile Lys Ala Tyr Gly Ala Lys Ile Asp Gly Val Gly Leu Gln Gly His
 195 200 205
 Phe Ile Leu Gly Ser Thr Pro Ser Gln Ser Ala Leu Thr Ser Val Leu
 210 215 220
 Glu Gly Tyr Thr Ala Leu Gly Val Glu Val Ala Tyr Thr Glu Leu Asp
 225 230 235 240
 Ile Arg Met Gln Leu Pro Ser Thr Asp Ala Lys Leu Ala Gln Gln Ala
 245 250 255
 Thr Glu Tyr Gln Asn Val Ala Ala Ala Cys Val Gly Thr Thr Gly Cys
 260 265 270
 Val Gly Val Thr Ile Trp Asp Trp Thr Asp Lys Tyr Ser Trp Val Pro
 275 280 285
 Ser Val Phe Ser Gly Tyr Gly Ala Ala Leu Pro Trp Asp Glu Asn Tyr
 290 295 300
 Val Lys Lys Pro Ala Tyr Asn Gly Ile Leu Ala Gly Leu Gly Ala Ser
 305 310 315 320
 Gly Ser Thr Met Thr Thr Thr Leu Thr Thr Thr Lys Thr Ser Thr
 325 330 335
 Thr Thr Thr Ala Thr Ser Thr Gly Val Ala Gln Lys Trp Gly Gln Cys
 340 345 350
 Gly Gly Ile Gly Trp Thr Gly Pro Thr Thr Cys Val Ser Gly Thr Thr
 355 360 365
 Cys Gln Lys Leu Asn Asp Trp Tyr Ser Gln Cys Leu
 370 375 380

<210> SEQ ID NO 4

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<211> LENGTH: 45
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Artificial DNA Primer
 <400> SEQUENCE: 4
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45

<210> SEQ ID NO 5
 <211> LENGTH: 40
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Artificial DNA Primer
 <400> SEQUENCE: 5
 ccctctagat ctgcggttac agacactgcg agtaccaatc

40

What is claimed is:

1. A process for degrading a cellulosic or hemicellulosic material, comprising: pretreating the cellulosic or hemicellulosic material with steam, dilute acid, hot water, alkaline pretreatment, lime, wet oxidation, wet explosion, ammonia fiber explosion, organosolve, ammonia percolation, ultrasound, electroporation, microwave, supercritical CO₂, supercritical H₂O, ozone, ionic liquid, or gamma irradiation; treating the pretreated cellulosic or hemicellulosic material with an enzyme composition comprising an isolated polypeptide having at least 99% or 100% sequence identity to the mature polypeptide of SEQ ID NO: 2, wherein the polypeptide has xylanase activity; and recovering the degraded cellulosic or hemicellulosic material.

2. The process of claim 1, wherein the degraded cellulosic or hemicellulosic material is a sugar.

3. The process of claim 2, wherein the sugar is selected from the group consisting of glucose, xylose, mannose, galactose, and arabinose.

4. A process for producing a fermentation product, comprising: (a) saccharifying a cellulosic or hemicellulosic material with an enzyme composition comprising a polypeptide having at least 99% or 100% sequence identity to the mature polypeptide of SEQ ID NO: 2, wherein the polypeptide has xylanase activity; (b) fermenting the saccharified cellulosic or hemicellulosic material with one or more fermenting microorganisms to produce the fermentation product; and (c) recovering the fermentation product from the fermentation.

5. The process of claim 4, wherein steps (a) and (b) are performed simultaneously in a simultaneous saccharification and fermentation.

6. A process of fermenting a cellulosic or hemicellulosic material, comprising: fermenting the cellulosic or hemicellulosic material with one or more fermenting microorganisms, wherein the cellulosic or hemicellulosic material is saccharified with an enzyme composition comprising a polypeptide having at least 99% or 100% sequence identity to the mature polypeptide of SEQ ID NO: 2, wherein the polypeptide has xylanase activity, wherein the fermenting of the cellulosic or hemicellulosic material produces a fermentation product; and recovering the fermentation product from the fermentation.

7. The process of claim 1, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

8. The process of claim 4, wherein the fermentation product is an alcohol, an alkane, a cycloalkane, an alkene, an amino acid, a gas, isoprene, a ketone, an organic acid, or polyketide.

9. The process of claim 4, wherein the cellulosic or hemicellulosic material is pretreated before saccharification.

10. The process of claim 4, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

11. The process of claim 6, wherein the cellulosic or hemicellulosic material is pretreated before saccharification.

12. The process of claim 6, wherein the enzyme composition further comprises one or more enzymes selected from the group consisting of a cellulase, an AA9 polypeptide, a hemicellulase, a cellulose inducible protein, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

13. The process of claim 1, further comprising, prior to pretreating, reducing the particle size, sieving, pre-soaking, wetting, washing, and/or conditioning the cellulosic or hemicellulosic material.

14. The process of claim 1, wherein the pretreatment occurs in an aqueous slurry and the cellulosic or hemicellulosic material is present during pretreatment in an amount between 10-80 wt %.

15. The process of claim 14, wherein the cellulosic or hemicellulosic material is present during pretreatment in an amount between 20-70 wt %.

16. The process of claim 14, wherein the cellulosic or hemicellulosic material is present during pretreatment in an amount between 30-60 wt %.

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